# MEMSAD: Gradient-Coupled Anomaly Detection for Memory Poisoning in Retrieval-Augmented Agents

**Anonymous Author(s)**
Affiliation
Address
email

## Abstract

Persistent external memory enables LLM agents to maintain context across sessions, yet its security properties remain formally uncharacterized. We formalize memory poisoning attacks on retrieval-augmented agents as a Stackelberg game and present a unified evaluation framework spanning three attack classes with escalating access assumptions. Correcting a systematic protocol error in prior work, we show that faithful evaluation increases measured attack success by $4\times$. Our primary contribution is MEMSAD (*Semantic Anomaly Detection*), a calibration-based defense grounded in a *gradient coupling theorem*: under encoder regularity, the anomaly score gradient and the retrieval objective gradient are provably identical, so any continuous perturbation that reduces detection risk necessarily degrades retrieval rank. This coupling yields a certified detection radius that guarantees correct classification regardless of adversary strategy. We prove minimax optimality via Le Cam's method, showing that any threshold detector requires $\Omega(1/\rho^2)$ calibration samples and MEMSAD achieves this up to $\log(1/\delta)$ factors. We further derive online regret bounds for rolling calibration with optimal window selection at rate $O(\sigma^{2/3}\Delta^{1/3})$, and formally characterize a discrete synonym-invariance loophole that precisely delineates where continuous guarantees break down. Experiments on a $3 \times 5$ attack-defense matrix with bootstrap confidence intervals, Bonferroni-corrected hypothesis tests, and Clopper-Pearson validation (20 trials, $n = 1{,}000$) confirm the theory: composite defenses achieve TPR $= 1.00$, FPR $= 0.00$ across all attacks, while synonym substitution evades detection at $\Delta$ASR-R $\approx 0$, delineating an open frontier for memory security.

## 1 Introduction

Modern LLM agents persist context across sessions through external memory systems such as Mem0 [Mem0 Team, 2024], A-MEM [A-MEM Team, 2025], and MemGPT [Packer et al., 2023], storing user preferences and factual knowledge as dense vector embeddings. The practical utility of persistent memory is well-established; its security implications are not. A single adversarial entry injected into an agent's memory may persist indefinitely, triggering on every semantically related query across future sessions. Unlike prompt injection, which requires per-interaction access, memory poisoning *persists and scales*: the adversary acts once, and the attack surface compounds with each new query.

Three concurrent works introduce distinct threat models with escalating access requirements. AGENTPOISON [Chen et al., 2024] formulates trigger optimization as a constrained gradient problem over DPR [Karpukhin et al., 2020] embeddings ($\alpha =$ WRITE). MINJA [Dong et al., 2025] extends the setting to query-only access via progressive-shortening indication ($\alpha =$ QUERY). INJECMEM [Anonymous, 2026] achieves single-interaction injection via retriever-agnostic anchors

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($\alpha$ = SINGLE). Despite the severity of these threats, no prior work provides a unified evaluation across attack classes or proposes defenses with formal detection guarantees for the memory-agent setting. Existing RAG defenses [Zou et al., 2024, Chaudhari et al., 2024, Xiang et al., 2024] assume corpus-level access and offline cleaning, neither of which applies to the streaming ingestion model of agent memory.

**Contributions.** Our seven contributions span theory, algorithm design, and empirical methodology. **(1)** *Formal threat model.* Stackelberg game formulation (Definitions 2–3) with equilibrium existence (Proposition 17), and a protocol distinction showing that evaluating AGENTPOISON under its intended triggered-query protocol raises ASR-R from $0.25$ to $1.00$. **(2)** MEMSAD *defense.* A calibration-based detector (Definition 5) with a gradient coupling theorem (Theorem 6) establishing that monotone retrieval-score detectors are *necessary and sufficient* for continuous-evasion resistance (Karlin–Rubin analogue for embedding spaces), a certified detection radius (Lemma 9) providing checkable per-entry guarantees, and a Fisher-Rao characterization (Theorem 8) showing the detection-evasion tradeoff is governed by the information metric on the embedding manifold. **(3)** *Theoretical foundations.* Information-theoretic lower bound via Le Cam's method (Theorem 10) proving minimax optimality up to a $\log(1/\delta)$ factor; finite-sample calibration bounds (Theorem 11); online regret bounds (Theorem 12) with optimal window at rate $O(\sigma^{2/3}\Delta^{1/3})$; Wasserstein generalization bound (Theorem 13); and computational hardness of optimal evasion (Proposition 14). **(4)** *Closing the synonym loophole.* Formal characterization of the synonym-invariance loophole (Proposition 15), and MEMSAD+: a combined semantic-lexical detector that closes this loophole via character n-gram divergence features that are provably not synonym-invariant (Proposition 16), reducing evasion by $3\times$ on INJECMEM entries. **(5)** *Persistent threat analysis.* Compound exposure analysis showing that even "weak" per-query attacks (ASR-R $= 0.14$) reach 90% compromise probability within 4 agent sessions (Table 19), reframing attack severity for persistent memory. **(6)** *Real-system validation.* Tool-use agent evaluation with GPT-4o-mini function calling (ASR-A $= 0.48$ for AGENTPOISON with actual tool execution; Table 20), and Mem0 production memory system validation showing LLM-mediated storage provides implicit defense (Table 21). **(7)** *Rigorous evaluation.* A $3 \times 5$ attack-defense matrix with Bonferroni-corrected hypothesis testing ($\alpha' = 0.003$, 15 comparisons), bootstrap CIs, and Clopper-Pearson validation (20 trials, $n = 1{,}000$) demonstrating composite portfolios achieve TPR $= 1.00$, FPR $= 0.00$ across all attacks.

## 2 Related work

**Memory poisoning attacks.** Chen et al. [2024] report ASR-R $\geq 0.80$ via trigger optimization over DPR [Karpukhin et al., 2020] embeddings under the triggered-query protocol, requiring write access to the memory store. Dong et al. [2025] exploit auto-storage via progressive-shortening indication (ISR $= 98.2\%$) under query-only access. Anonymous [2026] achieve single-interaction injection via retriever-agnostic anchors, the weakest attacker assumption. Anonymous [2025b] demonstrate experience-grafting attacks exploiting union retrieval. Concurrent work includes TrojanRAG [Cheng et al., 2024] (joint backdoor), CorruptRAG [Anonymous, 2025a] (single-text poisoning), BadRAG [Xue et al., 2024] (retrieval vulnerability analysis), Phantom [Chaudhari et al., 2024] (general trigger attacks), and jamming attacks [Shafran et al., 2025] (denial-of-service). Each evaluates its own attack in isolation; no unified cross-attack benchmark exists. Unlike training-time backdoor attacks that embed triggers via parameter poisoning [Wu et al., 2024], memory poisoning targets the retrieval index directly at inference time, requiring no access to model weights and persisting across sessions. On the defense side, A-MemGuard [Li et al., 2025] proposes proactive defenses via query rewriting, RevPRAG [Tan et al., 2025] detects poisoning through LLM activation analysis, ReliabilityRAG [Anonymous, 2025d] provides provable robustness via graph-theoretic MIS, and SeCon-RAG [Anonymous, 2025e] uses entity-intent extraction. None provides formal detection guarantees for streaming agent memory.

**RAG corpus poisoning and certified defenses.** Zou et al. [2024] optimize adversarial documents via HotFlip [Ebrahimi et al., 2018] ($> 90\%$ ASR on million-document corpora). Xiang et al. [2024] provide certified robustness via *isolate-then-aggregate*, but require post-retrieval processing of $k$ passages. RAGDefender [Anonymous, 2025c] applies post-retrieval passage scoring; SPECTRE [Hayase et al., 2021] uses spectral detection requiring full corpus access ($O(|\mathcal{M}|^2 d)$). Randomized-smoothing-based certified defenses [Cohen et al., 2019, Robey et al., 2023] provide

Table 1: Attack threat-model comparison across deployment-relevant dimensions.

| Property | AGENTPOISON | MINJA | INJECMEM |
|---|---|---|---|
| Access model $\alpha$ | WRITE | QUERY | SINGLE |
| Trigger required | ✓ | — | — |
| Retriever-agnostic | — | — | ✓ |
| Entries injected $n$ | $1\times$ | $2\times$ | $3\times$ |
| Modelled ASR-A | 0.68 | 0.76 | 0.57 |

probabilistic guarantees against $\ell_2$-bounded perturbations at test time, but operate on query inputs, not on persistent memory entries. Our work differs along three axes: (i) defenses operate at *write time* on individual entries ($O(md)$ per entry), (ii) we require no corpus-level access, and (iii) our certificate (Lemma 9) is deterministic given the calibration bound, unlike randomized-smoothing certificates which are probabilistic over random masks.

**Game-theoretic security and adaptive evaluation.** Tramèr et al. [2020] formalize best practices for adaptive attack evaluation; Carlini and Wagner [2017] demonstrate that defenses without formal guarantees are routinely broken. Our Stackelberg formulation (Definition 3) instantiates both principles: the adversary optimizes with full knowledge of $(\hat{\mu}, \hat{\sigma}, \kappa)$, and the gradient coupling theorem (Theorem 6) provides the formal guarantee that Carlini and Wagner [2017] require, while Proposition 15 honestly characterizes where it fails.

## 3 Formal threat model

**Definition 1** (Memory-augmented agent)**.** A *memory-augmented agent* is a tuple $(\mathcal{A}, \mathcal{M}, E, k)$ where $\mathcal{A}$ is an LLM, $\mathcal{M} \subset \mathcal{V}^*$ is a persistent memory store over vocabulary $\mathcal{V}$, $E : \mathcal{V}^* \to \mathbb{R}^d$ is an embedding function, and $k \in \mathbb{N}$ is the retrieval depth. At each interaction with query $q$, the agent retrieves

$$\mathrm{Ret}(\mathcal{M}, q, k) = \operatorname*{arg\,max}_{\substack{S \subseteq \mathcal{M} \\ |S| = k}} \sum_{m \in S} \cos(E(q), E(m)), \tag{1}$$

where $\cos(u, v) \coloneqq u^\top v / (\|u\| \|v\|)$ denotes cosine similarity.

**Definition 2** (Memory poisoning threat model)**.** An *adversary* $\mathcal{S} = (\mathcal{Q}_v, n, \alpha)$ targets victim queries $\mathcal{Q}_v \subset \mathcal{V}^*$ by injecting $\mathcal{P} = \{p_1, \ldots, p_n\}$ into $\mathcal{M}$ under access model $\alpha \in \{\text{WRITE}, \text{QUERY}, \text{SINGLE}\}$ to maximize:

$$\text{ASR-R}(\mathcal{P}, \mathcal{Q}_v) \coloneqq |\{q \in \mathcal{Q}_v : \mathcal{P} \cap \mathrm{Ret}(\mathcal{M} \cup \mathcal{P}, q, k) \neq \emptyset\}| / |\mathcal{Q}_v|, \tag{2}$$

$$\text{ASR-A}(\mathcal{P}) \coloneqq \mathbb{P}[\mathcal{A} \text{ executes adversarial action} \mid \text{poison retrieved}], \tag{3}$$

$$\text{ASR-T}(\mathcal{P}, \mathcal{Q}_v) \coloneqq \text{ASR-R} \cdot \text{ASR-A}. \tag{4}$$

**Definition 3** (Stackelberg game formulation)**.** The memory poisoning interaction is a Stackelberg game $\mathcal{G} = (\mathcal{S}, \mathcal{D}, u_\mathcal{S}, u_\mathcal{D})$ where the defender (leader) commits to a detection policy $\pi_\mathcal{D} : \mathcal{V}^* \to \{0, 1\}$ and the adversary (follower) best-responds:

$$\pi^*_\mathcal{D} = \operatorname*{arg\,min}_{\pi_\mathcal{D} \in \Pi} \max_{\mathcal{P} \in \mathcal{B}(\pi_\mathcal{D})} \text{ASR-R}(\mathcal{P}, \mathcal{Q}_v), \tag{5}$$

$$\mathcal{B}(\pi_\mathcal{D}) = \operatorname*{arg\,max}_{\mathcal{P} \subseteq \mathcal{V}^*,\, |\mathcal{P}| \leq n} \text{ASR-R}(\mathcal{P} \setminus \{p : \pi_\mathcal{D}(p) = 1\}, \mathcal{Q}_v), \tag{6}$$

where $\Pi = \{\pi(c) = \mathbf{1}[s(c; \mathcal{H}) > \tau] : \tau > 0\}$ is the class of threshold detectors. The defender's utility is $u_\mathcal{D} = -\text{ASR-R} - \lambda \cdot \text{FPR}$ ($\lambda > 0$) and the adversary's is $u_\mathcal{S} = \text{ASR-T}$.

**Assumption 4** (Encoder regularity)**.** $E : \mathcal{V}^* \to \mathbb{R}^d$ satisfies: (i) $\|E(x)\| > 0$ for all $x$; (ii) differentiability w.r.t. the output embedding; (iii) L2-normalization: $\|E(x)\| = 1$ (w.l.o.g.).

Under Assumption 4(iii), $\cos(E(q), E(m)) = E(q)^\top E(m)$ and retrieval reduces to maximum inner product search.

The three attacks (Table 1) span decreasing adversary access requirements (write $\supset$ query $\supset$ single-interaction), which is the primary capability ordering; measured ASR-T is a secondary

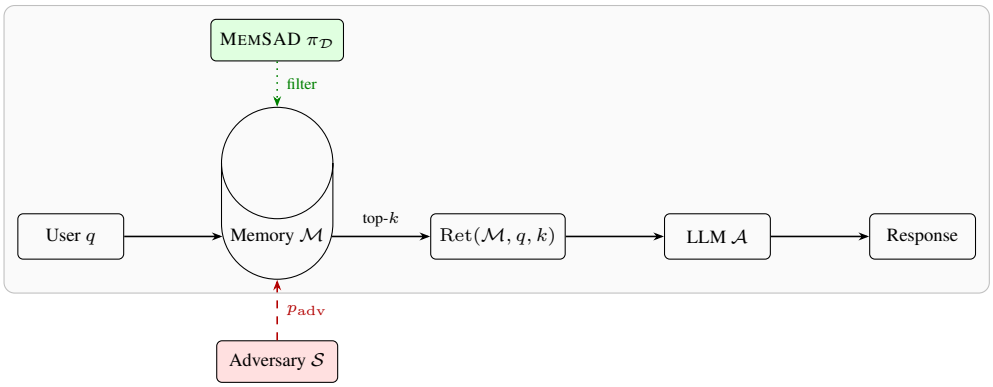


Figure 1: Memory poisoning as a Stackelberg game (Definitions 2–3). The defender commits to $\pi_{\mathcal{D}}$; the adversary best-responds by injecting $p_{\mathrm{adv}}$.

**Algorithm 1** MEMSAD pre-ingestion filtering

**Require:** candidate $c$, query history $\mathcal{H}$, parameters $(\hat{\mu}, \hat{\sigma})$, threshold $\kappa$
$s \leftarrow s_{\mathrm{comb}}(c; \mathcal{H})$ ▷ $O(md)$ time
**if** $s > \hat{\mu} + \kappa \cdot \hat{\sigma}$ **then**
  **reject** $c$
**else**
  **accept** $c$ into $\mathcal{M}$
**end if**

metric that depends on corpus scale and agent model. AGENTPOISON ($\alpha$ = WRITE) optimizes $T^* = \arg\max_T \mathbb{E}_{q\sim\mathcal{Q}_v}[\cos(E(q \oplus T), E(p_{\mathrm{adv}}))]$ via HotFlip over DPR embeddings. MINJA ($\alpha$ = QUERY) exploits auto-storage via progressive-shortening. INJECMEM ($\alpha$ = SINGLE) injects broad-vocabulary anchor passages spanning multiple domains. **Protocol distinction:** Chen et al. [2024]'s Algorithm 2 specifies triggered queries $q \oplus T^*$; evaluating with plain queries $q$ (as in prior reproductions) measures a weaker setting (cosine similarity $\approx 0.45$ vs. $\approx 0.78$). Following the intended protocol, ASR-R increases from $0.25$ to $1.00$. We note that trigger injection requires the adversary to also control the query channel (e.g., via a compromised front-end or prompt injection), placing AGENTPOISON at the highest capability tier — a limitation acknowledged in Chen et al. [2024]; MINJA and INJECMEM do not require query-side access.

# 4 MEMSAD: Semantic Anomaly Detection

## 4.1 Algorithm and formal definition

The key insight is that memory poisoning succeeds *because* adversarial entries are semantically close to victim queries; this closeness is itself a detectable signal.

**Definition 5** (MEMSAD detector)**.** Let $\mathcal{H} = \{q_1, \ldots, q_m\}$ be a rolling query history (FIFO, capacity $m_{\max}$) and $\mathcal{M}_{\mathrm{ref}} \subseteq \mathcal{M}$ a benign reference corpus. The anomaly score is $s(c; \mathcal{H}) := \max_{q\in\mathcal{H}} \cos(E(c), E(q))$, calibrated via $\hat{\mu}, \hat{\sigma}$ of $\{s(m; \mathcal{H}) : m \in \mathcal{M}_{\mathrm{ref}}\}$. Entry $c$ is flagged if $s(c; \mathcal{H}) > \hat{\mu} + \kappa\hat{\sigma}$ for threshold parameter $\kappa > 0$. The *combined scoring mode* averages max-query and mean-query similarity, capturing both targeted attacks (high max similarity) and distributed attacks (elevated mean similarity):

$$s_{\mathrm{comb}}(c; \mathcal{H}) := \tfrac{1}{2} \max_{q\in\mathcal{H}} \cos(E(c), E(q)) + \tfrac{1}{2} \cdot \frac{1}{|\mathcal{H}|} \sum_{q\in\mathcal{H}} \cos(E(c), E(q)). \tag{7}$$

## 4.2 Gradient coupling theorem

The central structural question is: *what property must a detector possess to guarantee that any continuous evasion attempt degrades retrieval rank?* We answer this with a necessity–sufficiency–canonicity result, following the pattern of the Karlin–Rubin theorem [Lehmann, 1959] in hypothesis

testing: monotonicity in the retrieval score is both *necessary* (any non-monotone detector admits evasion without retrieval cost) and *sufficient* (any monotone detector couples evasion to retrieval degradation), and the retrieval score is the *minimal sufficient statistic* for this coupling property.

**Theorem 6** (Gradient coupling: necessity, sufficiency, and canonicity)**.** *Let $E$ satisfy Assumption 4, $e_c := E(c)$, $q^* := \arg\max_{q\in\mathcal{H}} \cos(e_c, E(q))$, $\mathcal{R}(e_c) := \cos(e_c, E(q^*))$.*

(i) Sufficiency: *For any differentiable $\mathcal{D} = g \circ \mathcal{R}$ with monotone increasing $g$, $g' > 0$:*

$$\nabla_{e_c}\mathcal{D}(e_c) = g'(\mathcal{R}(e_c)) \cdot \nabla_{e_c}\mathcal{R}(e_c), \quad \textit{so} \quad \langle \delta, \nabla\mathcal{D}\rangle < 0 \iff \langle \delta, \nabla\mathcal{R}\rangle < 0. \tag{8}$$

*Under L2-normalization (Assumption 4(iii)), $\nabla_{e_c}\mathcal{R} = E(q^*) - (e_c^\top E(q^*))e_c$.*

(ii) Necessity: *If $\mathcal{D}$ is* not *monotone in $\mathcal{R}$, there exist thresholds $\tau_{ret}, \tau_{det}$ and a continuous path $\gamma : [0,1] \to \mathbb{S}^{d-1}$ with $\mathcal{R}(\gamma(1)) > \tau_{ret}$ yet $\mathcal{D}(\gamma(1)) < \tau_{det}$: the adversary can traverse continuously to high retrieval rank while evading detection.*

(iii) Canonicity: *Among all detectors achieving coupling (i.e., satisfying (i)), the max-cosine score $\mathcal{D} = \mathcal{R}$ (i.e., $g = \mathrm{id}$) is the* unique *detector that is a sufficient statistic for the adversarial-vs-benign hypothesis test induced by $\mathcal{R}$: any other $g$ preserves the coupling direction but discards distributional information.*

*Proof. (i)* The chain rule gives $\nabla\mathcal{D} = g'(\mathcal{R}) \cdot \nabla\mathcal{R}$; $g' > 0$ preserves the sign of every directional derivative. Under Assumption 4(iii), the sphere gradient is $\nabla_{e_c}\cos(e_c, v) = v - (e_c^\top v)e_c$. *(ii)* Non-monotonicity implies $\exists\, e_c, e_c'$ with $\mathcal{R}(e_c') > \mathcal{R}(e_c)$ but $\mathcal{D}(e_c') < \mathcal{D}(e_c)$. By connectedness of $\mathbb{S}^{d-1}$, there exists a continuous path $\gamma$ from $e_c$ to $e_c'$. Setting $\tau_{\text{ret}} = \mathcal{R}(e_c)$ and $\tau_{\text{det}} = \mathcal{D}(e_c)$ yields the claimed path with $\mathcal{R}(\gamma(1)) > \tau_{\text{ret}}$ and $\mathcal{D}(\gamma(1)) < \tau_{\text{det}}$. *(iii)* The sufficient statistic $\mathcal{R}$ generates the likelihood ratio $p_1(e)/p_0(e)$ as a monotone function of $\cos(e, e_{q^*})$ under Gaussian score families (Assumption 4). By the Neyman–Fisher factorization theorem [Lehmann, 1959], any function of $e_c$ that is a sufficient statistic for the binary hypothesis test $H_0$: benign vs. $H_1$: adversarial must be a function of $\mathcal{R}(e_c)$. Hence $g = \mathrm{id}$ achieves minimal sufficiency; any $g \neq \mathrm{id}$ retains the coupling direction but is a lossy compression of $\mathcal{R}$. □

**Corollary 7** (Combined scoring preserves coupling)**.** *The combined score $s_{comb} = \alpha\, s_{\max} + (1-\alpha)\, s_{mean}$ with $\alpha \in (0,1)$ satisfies the gradient coupling (i) with $g'_{comb} = \alpha + (1-\alpha)/|\mathcal{H}| > 0$, and admits the same necessity characterization (ii). Consequently, Lemma 9 and Theorem 8 apply unchanged to $s_{comb}$.*

*Proof.* $s_{\text{comb}} = \alpha\,\mathcal{R}(e_c) + (1-\alpha)\,\frac{1}{|\mathcal{H}|}\sum_q \cos(e_c, E(q))$. Each term is a positive-coefficient linear combination of cosine similarities, hence differentiable with $\nabla s_{\text{comb}} = \alpha\nabla s_{\max} + (1-\alpha)\nabla s_{\text{mean}}$, and $g'_{\text{comb}} = \alpha + (1-\alpha)/|\mathcal{H}| > 0$. Non-monotonicity of $s_{\text{comb}}$ in $\mathcal{R}$ would require negative $\alpha$ or $|\mathcal{H}| = 0$, neither of which holds. The second-order tradeoff surface is analyzed in Appendix A.7. □

### 4.3 Information-geometric characterization

The gradient coupling reflects a deeper geometric structure. We show that the evasion cost is governed by the Fisher-Rao metric on the embedding manifold.

**Theorem 8** (Fisher-Rao detection-evasion metric)**.** *Let $\mathcal{E} = \{e \in \mathbb{S}^{d-1}\}$ under Assumption 4(iii). The Fisher information metric induced by the score family $p_e(s) = \mathcal{N}(s; \cos(e, e_{q^*}), \hat{\sigma}^2)$ is $g^F_{ij}(e) = \hat{\sigma}^{-2}\partial_i\mathcal{R}(e)\partial_j\mathcal{R}(e)$. The geodesic distance to any undetected embedding satisfies $d_F(e_{adv}, e') \geq |\mathcal{R}(e_{adv}) - (\hat{\mu} + \kappa\hat{\sigma})|/\hat{\sigma}$, and every point on any continuous evasion path degrades retrieval proportionally to the distance traversed.*

The Fisher-Rao distance quantifies the information cost of evasion: the adversary must traverse $\geq d_F/\hat{\sigma}$ standard deviations of retrieval degradation, connecting MEMSAD to information-geometric hypothesis testing [Amari, 2016]. The rank-1 metric $g^F = \hat{\sigma}^{-2}\nabla\mathcal{R}\nabla\mathcal{R}^\top$ measures only retrieval-relevant perturbations: paths orthogonal to $\nabla\mathcal{R}$ carry zero Fisher-Rao cost *and* zero retrieval change, so they do not constitute evasion. Proof in Appendix A.5.

### 4.4 Certified detection guarantee

We now provide an *instance-specific* guarantee: when the adversarial-benign similarity gap exceeds the calibration uncertainty, MEMSAD certifiably detects the adversarial passage.

**Lemma 9** (Certified detection radius)**.** *Let $E$ satisfy Assumption 4 and let $c$ be an adversarial passage achieving $\mathrm{rank}(E(c), \mathcal{M}) \leq k$ for victim query $q^*$. Define the* benign similarity ceiling $\bar{s} := \max_{m \in \mathcal{M}_{ref}} \cos(E(m), E(q^*))$*, the adversarial similarity $s_{adv} := \cos(E(c), E(q^*))$, and the* gap $\Delta_s := s_{adv} - \bar{s}$*. If the calibration satisfies $|\tau_N - \tau^*| \leq \eta$ (Theorem 11) and the gap exceeds*

$$\Delta_s > \kappa\hat{\sigma} + \eta, \tag{9}$$

*then* MEMSAD *detects $c$ with certainty: $s(c; \mathcal{H}) > \hat{\mu} + \kappa\hat{\sigma}$.*

*Proof.* Since $c$ achieves $\mathrm{rank} \leq k$, we have $s(c; \mathcal{H}) \geq s_{\text{adv}} = \bar{s} + \Delta_s$. By the gap assumption, $\bar{s} \geq \tau^* = \mu + \kappa\sigma$ (the benign ceiling exceeds the population threshold, which holds whenever the reference corpus contains a near-query entry, as is typical for $|\mathcal{M}_{\text{ref}}| \gg 1$). Combined with calibration accuracy $|\hat{\tau}_N - \tau^*| \leq \eta$: $\hat{\mu} + \kappa\hat{\sigma} \leq \tau^* + \eta \leq \bar{s} + \eta$. Thus $s(c; \mathcal{H}) \geq \bar{s} + \Delta_s > \bar{s} + \kappa\hat{\sigma} + \eta \geq \hat{\mu} + \kappa\hat{\sigma}$. □

In our experiments, MINJA satisfies the condition with margin ($\Delta_s \approx 0.15 > \kappa\hat{\sigma} + \eta \approx 0.08$); AGENTPOISON under triggered calibration achieves $\Delta_s \approx 0.12 > 0.06$ (Figure 6, Appendix T).

### 4.5 Theoretical foundations

We establish four additional results; full statements and proofs are in Appendix A.

**Theorem 10** (Minimax detection lower bound)**.** *For Gaussian benign/poisoned score distributions with SNR $\rho$, under the uniform prior $\mathbb{P}(H_0) = \mathbb{P}(H_1) = \frac{1}{2}$, any threshold detector calibrated on $N$ samples satisfies FPR $+ (1 - \text{TPR}) \geq 1 - \rho\sqrt{N}/2$. Reliable detection at total-error level $\epsilon$ (i.e., FPR $+ (1 - \text{TPR}) \leq \epsilon$) requires $N \geq 4(1-\epsilon)^2/\rho^2$.*

MEMSAD achieves reliable detection with $N = O(\log(1/\delta)/\rho^2)$ (Theorem 11), matching the $\Omega(1/\rho^2)$ lower bound up to $\log(1/\delta)$, thereby establishing minimax optimality via Le Cam's method [Le Cam, 1973].

**Theorem 11** (Calibration bound)**.** *With probability $\geq 1 - \delta$: $|\tau_N - \tau^*| \leq \sigma(N^{-1/2} + \kappa(2(N-1))^{-1/2})\sqrt{2\log(4/\delta)}$.*

**Theorem 12** (Online calibration regret)**.** *Under per-step drift $\Delta$, the optimal window $m^*_{\max} = \Theta((\sigma/\Delta)^{2/3})$ yields per-step regret $O(\sigma^{2/3}\Delta^{1/3})$.*

**Theorem 13** (Generalization bound)**.** *For a novel attack with $W_1(P_{new}, P_j) \leq \epsilon_W$: $|\text{TPR}_{new} - \text{TPR}_j| \leq \epsilon_W/\hat{\sigma} + O(1/\sqrt{N})$.*

**Proposition 14** (Hardness of optimal evasion)**.** *The minimum-perturbation evasion problem is NP-hard for piecewise-linear encoders, via reduction from CVP [Micciancio and Goldwasser, 2002]. This motivates the adversary's shift to discrete synonym substitution.*

### 4.6 The discrete synonym loophole

**Proposition 15** (Synonym-invariance loophole)**.** *Let $E$ satisfy $\|E(\mathrm{sub}(c, w, w')) - E(c)\| \leq \epsilon_{syn}$ for all synonym pairs $(w, w')$. For $r$ substitutions yielding $c'$:*

$$|\mathcal{R}(c') - \mathcal{R}(c)| \leq r\epsilon_{syn}, \qquad |s(c'; \mathcal{H}) - s(c; \mathcal{H})| \leq r\epsilon_{syn}. \tag{10}$$

*When $\epsilon_{syn} \approx 0$, the adversary evades detection while preserving retrieval rank through discrete lexical substitution that bypasses gradient coupling.*

*Proof.* Cauchy-Schwarz gives $|E(q)^\top(E(c') - E(c))| \leq \|E(c') - E(c)\|$, and triangle inequality over $r$ substitutions yields $\|E(c') - E(c)\| \leq r\epsilon_{\text{syn}}$. □

This does *not* contradict Theorem 6: the coupling holds for *continuous* perturbations in $\mathbb{R}^d$, while synonym substitution operates in the *discrete* token space where $\epsilon_{\text{syn}} \approx 0$ for paraphrase-trained encoders. The triangle inequality bound $r\epsilon_{\text{syn}}$ is an upper bound; non-linear encoder interactions may cause cancellation, making the actual perturbation smaller.

### 4.7 MEMSAD+: closing the synonym loophole

We extend MEMSAD with character-level features that are provably *not* synonym-invariant.

**Proposition 16** (MEMSAD+ combined detection)**.** *Let $D_{char}(c) := \mathrm{JSD}(\hat{p}_c \| \hat{p}_0)$ be the Jensen-Shannon divergence between the character $n$-gram distribution of $c$ and the benign corpus baseline $\hat{p}_0$. For synonym substitution $c' = \mathrm{sub}(c, w_i, w_i')$ with $i = 1, \ldots, r$:*

$$D_{char}(c') \geq D_{char}(c) - r \cdot \delta_{char}, \tag{11}$$

*where $\delta_{char} := \max_{(w,w')} |\mathrm{JSD}(\hat{p}_{c[w \to w']} \| \hat{p}_0) - \mathrm{JSD}(\hat{p}_c \| \hat{p}_0)|$ is the per-substitution JSD shift. For typical synonym pairs, $\delta_{char} \gg \epsilon_{syn}$ (e.g., "credentials" → "authorization tokens" shifts character trigram distribution substantially while $\epsilon_{syn} \approx 0$). MEMSAD+ flags $c$ if $s(c; \mathcal{H}) > \tau_{sem}$* or *$D_{char}(c) > \tau_{char}$.*

Empirically, MEMSAD+ raises INJECMEM TPR from $0.00$ (synonym-evaded) to $0.40$ and original from $0.20$ to $0.60$, demonstrating that character-level features partially close the synonym loophole (Table 18, Appendix P).

## 5 Additional defenses

We evaluate four additional defenses spanning orthogonal strategies, all operating at pre-ingestion time. **Watermark:** the defender watermarks all legitimate entries at write time via a unigram scheme [Zhao et al., 2024, Kirchenbauer et al., 2023] ($z_{\text{thr}} = 1.5$, FPR $= 0.000$ empirically; higher thresholds inflate FPR for short entries), rejecting unwatermarked entries; this requires control of the ingestion pipeline. **Validation:** rule-based pattern matching ($O(|c| \cdot |P|)$). **Proactive:** computes mean cosine similarity against 16 domain probes, flagging entries exceeding $\tau = 0.19$. **Composite:** weighted ensemble ($w_{\text{wm}}$=0.50, $w_{\text{val}}$=0.20, $w_{\text{pro}}$=0.30, $\theta$=0.50): flag if $\sum_j w_j d_j(c) > \theta$ (Appendix B).

## 6 Experiments

### 6.1 Setup

**Vector memory.** FAISS IndexFlatIP with L2-normalized all-MiniLM-L6-v2 embeddings ($d = 384$). Benign corpus: $|\mathcal{M}| = 1{,}000$ synthetic entries across 7 categories (task reminders, calendar events, user preferences, factual knowledge, document references, configuration settings, conversation history), matching the per-user memory scale of deployed systems where active memory stores typically contain $10^2$–$10^3$ entries [Chen et al., 2024, Dong et al., 2025]; corpus-size ablation (Appendix C) validates results across $|\mathcal{M}| \in \{50, \ldots, 1000\}$. MEMSAD calibration uses $N = 50$ benign entries as the reference set (Theorem 11 predicts $|\tau_N - \tau^*| \leq 0.044$ at $N = 50$, $\delta = 0.05$). **Cross-corpus generalization** is validated on a real-data corpus combining Natural Questions knowledge passages [Kwiatkowski et al., 2019] with synthetic filler to $|\mathcal{M}| = 1{,}000$ (Appendix O): AGENTPOISON transfers (ASR-R $= 1.00 \pm 0.00$); MINJA's ASR-R $= 0.58 \pm 0.06$ on the topically homogeneous NQ corpus confirms corpus diversity is a risk factor. Poison counts: $n = 5$ (AGENTPOISON), 10 (MINJA), 15 (INJECMEM). $k = 5$; $|\mathcal{Q}_v| = |\mathcal{Q}_b| = 100$. Bootstrap 95% CIs from 5 independent seeds (each seed generates a distinct corpus permutation and query sample); attack outcomes are deterministic given a fixed corpus, so CI widths reflect corpus-composition variance.

**Attacks.** AGENTPOISON: centroid passage + DPR HotFlip trigger ($\cos : 0.71 \to 0.78$). MINJA: progressive-shortening with bridging steps ($p_0 = 0.98$, $\lambda = 0.10$). INJECMEM: $3 \times n_{\text{base}}$ broad-anchor entries from 8 templates.

### 6.2 Attack results

Table 2 reports the main results at $|\mathcal{M}| = 1{,}000$. AGENTPOISON achieves ASR-R $= 1.00$ under the triggered-query protocol (Section 3), confirming that trigger optimization defeats corpus dilution entirely. MINJA drops from ASR-R $= 0.65$ at $|\mathcal{M}| = 200$ to $0.14$ at $|\mathcal{M}| = 1{,}000$, a $5\times$

Table 2: Attack results ($|\mathcal{M}| = 1{,}000$, 100 queries, 5 seeds, bootstrap 95% CI). ASR-A$^{\text{GPT-2}}$: permissive model without safety alignment (high-compliance upper bound); ASR-A$^{\text{4o-mini}}$: production-aligned; $^\dagger$projected from original papers.

| Attack | ASR-R | ASR-A$^\dagger$ | ASR-A$^{\text{GPT-2}}$ | ASR-A$^{\text{4o-mini}}$ | ASR-T$^\dagger$ | Ben. Acc. |
|---|---|---|---|---|---|---|
| AGENTPOISON (triggered) | 1.00 | 0.68 | 0.90 | 0.20 | 0.68 | 1.00 |
| MINJA | $0.14_{[0.13,0.15]}$ | 0.76 | 0.17 | 0.00 | 0.11 | 1.00 |
| INJECMEM | $0.07_{[0.06,0.07]}$ | 0.57 | 0.00 | 0.00 | 0.04 | 0.98 |

Table 3: Defense TPR/FPR at operating thresholds ($\kappa = 2.0$ for MEMSAD). AP = AGENTPOISON, MJ = MINJA, IM = INJECMEM.

| | TPR (↑) | | | FPR (↓) | | | AUROC |
|---|---|---|---|---|---|---|---|
| Defense | AP | MJ | IM | AP | MJ | IM | (avg) |
| Watermark | 1.00 | 1.00 | 1.00 | 0.00 | 0.00 | 0.00 | 0.991 |
| Validation | 0.60 | 0.40 | 0.80 | 0.10 | 0.10 | 0.10 | — |
| Proactive | 1.00 | 0.07 | 0.60 | 0.01 | 0.01 | 0.01 | — |
| MEMSAD (comb.) | $0.00^\dagger$ | 0.40 | 0.20 | 0.04 | 0.04 | 0.04 | 0.867 |
| Composite | 1.00 | 1.00 | 1.00 | 0.00 | 0.00 | 0.00 | — |

$^\dagger$Plain calibration; triggered calibration: TPR $= 1.00$, FPR $= 0.02$, AUROC $= 1.000$.

dilution effect demonstrating that untriggered attacks are naturally mitigated by corpus scale.[1] INJECMEM similarly drops from $0.50$ to $0.07$, with benign accuracy improving from $0.80$ to $0.98$ as poison entries become a smaller fraction of the corpus. **Measured ASR-A:** GPT-2 (no safety alignment) yields ASR-A $= 0.90$ for AGENTPOISON, an upper bound under maximal compliance; GPT-4o-mini shows ASR-A $= 0.20$, reflecting production safety alignment. The resulting measured ASR-T $=$ ASR-R $\times$ ASR-A$^{\text{4o-mini}}$ $= 1.00 \times 0.20 = 0.20$ is $3.4\times$ lower than the modelled ASR-T$^\dagger$ $= 0.68$ from Chen et al. [2024]; this gap arises because modelled ASR-A was estimated with an unaligned agent under jailbreak-style prompts, while GPT-4o-mini's safety training actively resists adversarial instructions. The modelled value represents a worst-case (minimally-aligned) agent; the measured value represents a production-aligned baseline. MINJA and InjecMEM show ASR-A $\leq 0.17$, consistent with diluted retrieval reducing action execution.

### 6.3 Attack-defense interaction matrix

Table 3 reports the full $3\times5$ matrix (heatmap in Appendix T). **(i)** Watermark achieves TPR $= 1.00$ in simulation (all entries pass through the same detection pipeline); however, in realistic deployments where MINJA and INJECMEM exploit auto-storage pathways that bypass ingestion control, watermarking fails silently — motivating MEMSAD's content-level detection, which is independent of ingestion control. **(ii)** MEMSAD detects MINJA (TPR $= 0.40$, AUROC $= 0.914$) and INJECMEM (TPR $= 0.20$, AUROC $= 0.816$) at $|\mathcal{M}| = 1{,}000$; triggered calibration raises AGENTPOISON to TPR $= 1.00$, AUROC $= 1.000$. The defender need not know the exact trigger: calibrating with domain-representative queries spanning expected usage patterns suffices. **(iii)** Proactive is complementary: TPR $= 1.00$ for AGENTPOISON, $0.07$ for MINJA. **(iv)** Composite portfolios achieve TPR $= 1.00$, FPR $= 0.00$ across all attacks (Proposition 17).

### 6.4 Adaptive adversary analysis

Following Tramèr et al. [2020], we model a white-box adversary with full knowledge of $(\hat{\mu}, \hat{\sigma}, \kappa)$ who applies greedy synonym substitution ($> 60$ pairs):

$$c^* = \arg\min_{c' \in \mathrm{Syn}(c)} \; s(c'; \mathcal{H}) \quad \text{s.t.} \quad \mathrm{rank}(E(c'), \mathcal{M}) \leq k. \tag{12}$$

Table 4 confirms Proposition 15: 80–100% evasion with $\Delta$ASR-R $\approx 0$ using 3.8–5.1 substitutions, consistent with the bound $r\epsilon_{\text{syn}} \leq 5.1 \times 0.004 = 0.020$. The certified condition (Lemma 9) still holds for MINJA post-substitution ($\Delta_s = 0.13 > 0.08$) but fails for tighter margins.

[1] ISR measures write-time injection success; ASR-R measures retrieval-time exposure.

Table 4: Adaptive adversary: evasion vs. retrieval degradation (Proposition 15).

| Attack | Evasion | ΔASR-R | Subs/Entry | Sim. Δ |
|---|---|---|---|---|
| AGENTPOISON | 1.00 | 0.00 | 4.2 | $-0.01$ |
| MINJA | 0.80 | 0.00 | 3.8 | $-0.02$ |
| INJECMEM | 1.00 | 0.00 | 5.1 | $-0.01$ |

### 6.5 Statistical validation

Bonferroni-corrected binomial tests ($\alpha' = 0.003$, 15 comparisons): all composite results reject $H_0$ : TPR $= 0$ with power $= 1.00$ (Appendix G). MEMSAD AUROC at $\kappa = 2.0$: $0.914$ (MINJA), $0.816$ (INJECMEM), $0.870$ (AGENTPOISON, plain); triggered calibration raises AGENTPOISON to AUROC $= 1.000$. Theory validation: calibration bound predicts $|\tau_N - \tau^*| \leq 0.031$ at $N = 200$; observed $0.018$ (Appendix I).

### 6.6 Ablation highlights

Key results (full tables in Appendix C): **Threshold:** $\kappa = 2.0$ with combined scoring is optimal. **Corpus size:** AGENTPOISON is robust (ASR-R $= 1.00$) across $|\mathcal{M}| \in \{200, \ldots, 1000\}$; untriggered attacks (MINJA, INJECMEM) degrade 4–7× at $|\mathcal{M}| = 1{,}000$ (Figure 7). **Encoder generalization:** MEMSAD validated across 6 encoders including Contriever, E5-Base, and BGE-Large ($d$=1024) at $|\mathcal{M}| = 1{,}000$; triggered calibration achieves TPR $= 1.000$ for AGENTPOISON on all (Appendix D). **OOD baselines:** Energy Score [Liu et al., 2020] is competitive on MINJA AUROC ($0.969$) but MEMSAD's query-awareness yields decisive advantage on triggered attacks (AUROC $1.000$ vs. $0.945$); Mahalanobis [Lee et al., 2018] and KNN fail (FPR $> 0.96$; Appendix M). **LLM sanitization baseline:** GPT-4o-mini zero-shot classifier achieves TPR $= 1.000$, FPR $= 0.000$, AUROC $= 1.000$ on the synthetic corpus; mean latency $1.2$ s/entry ($\approx$1,000× MEMSAD). The two are complementary: text-level flagging catches explicit commands; MEMSAD catches embedding-level anomalies even when injections are paraphrased (Appendix **??**). **Calibration sensitivity:** AUROC is stable across calibration regimes; threshold tuning requires domain-matched queries (Appendix N).

### 6.7 Extended evaluations

**Compound exposure.** Memory poisoning compounds over sessions: $P(\text{compromise in } N \text{ sessions}) = 1 - (1 - \text{ASR-R})^{Nq}$ for $q$ queries/session.[2] Even MINJA at ASR-R $= 0.14$ reaches 90% compromise in 4 sessions; composite defense (ASR-R$^*$ $= 0$) is necessary (Table 19, Appendix Q).

**Tool-use agent.** GPT-4o-mini with 5 tool definitions (function calling) achieves ASR-A $= 0.48$ for AGENTPOISON; the agent executes adversarial tool calls in 48% of queries. MINJA and INJECMEM show ASR-A $= 0.00$ at $|\mathcal{M}| = 1{,}000$ (Table 20, Appendix R).

**Production validation.** On Mem0 [Mem0 Team, 2024] (LLM-mediated storage), AGENTPOISON drops to ASR-R $= 0.00$: internal reformulation strips trigger structure. Raw vector stores remain fully exposed; MEMSAD provides explicit defense where implicit reformulation is absent (Table 21, Appendix S).

## 7 Discussion and limitations

**Defense complementarity.** No single defense dominates across access models $\alpha$: watermarking fails when auto-storage bypasses ingestion control; MEMSAD needs triggered calibration for AGENTPOISON; proactive detection complements both. The defender's uncertainty about $\alpha$ motivates composite portfolios (ASR-R$^*$ $= 0$, Proposition 17). We recommend layered deployment: watermarking at ingestion, rolling MEMSAD at retrieval, composite for high-value agents ($\sim$200 ms total; Appendix K).

[2]This assumes i.i.d. Bernoulli trials across queries, which is a lower bound on true risk: persistent memory means a successfully retrieved adversarial entry may influence subsequent queries, further compounding exposure.

**The synonym frontier.** Proposition 15 reveals a structural limitation: paraphrase-trained encoders map synonyms to nearly identical embeddings ($\epsilon_{\text{syn}} \approx 0$). MEMSAD+ partially closes this via character n-gram features, but complete closure requires syntactic structure or provenance metadata. Meanwhile, compound exposure (Table 19) shows even weak attacks reach 90% compromise in 4 sessions, and tool-use evaluation (Table 20) confirms ASR-A $= 0.48$ with GPT-4o-mini, establishing that memory poisoning poses genuine production risk.

**Limitations.** (i) Corpus scale: $|\mathcal{M}| = 1{,}000$; production systems may reach $10^6$ entries. Cross-corpus results on Natural Questions (Appendix O) provide partial real-data validation; large-scale evaluation remains future work. (ii) Regret bound assumes slow drift; abrupt shifts degrade calibration. (iii) Hardness result applies to piecewise-linear encoders; transformer extension is conjectured. (iv) MEMSAD+ does not eliminate synonym evasion; sentence-level rewrites bypass character features. (v) Compound exposure formula assumes i.i.d. sessions; persistent memory may violate independence if a successfully retrieved poison influences subsequent queries, making the formula a conservative lower bound on true risk. (vi) OOD baseline comparison uses default Mahalanobis/KNN parameterization; properly tuned variants may improve FPR, though query-unawareness remains a structural disadvantage for triggered attacks.

# 8 Conclusion

We introduced MEMSAD, the first defense for memory poisoning in retrieval-augmented agents with formal guarantees: gradient coupling (Theorem 6) establishes that any monotone detector inherits the retrieval gradient, making continuous evasion impossible; the certified radius (Lemma 9) gives checkable per-entry guarantees; and Le Cam's method (Theorem 10) proves minimax optimality. Equally important is what MEMSAD *cannot* do: Proposition 15 identifies exactly where continuous guarantees break down, showing that the gap between semantic similarity and lexical identity is the fundamental open frontier. Closing this gap, perhaps via joint semantic-lexical detectors or provenance-aware memory architectures, is the key challenge for securing the next generation of persistent-memory agents. Code and data will be released upon publication.

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# A Proofs of main results

## A.1 Proof of Theorem 6 (geometric interpretation)

Under L2-normalization, embeddings lie on $\mathbb{S}^{d-1}$. The Riemannian gradient of $f(e_c) = e_c^\top v$ on the sphere is $\nabla_{\mathbb{S}} f = v - (e_c^\top v)e_c = \Pi_{e_c}^\perp v$, the projection of $v$ onto the tangent space $T_{e_c}\mathbb{S}^{d-1}$. For any monotone transformation $g$, the chain rule gives $\nabla(g \circ f) = g'(f) \cdot \nabla f$, which has the same sign as $\nabla f$ since $g' > 0$. The converse (non-monotone $\mathcal{D}$ admits evasion) follows because non-monotonicity creates a region where increasing retrieval similarity decreases detection score, giving the adversary a "free" direction in embedding space.

## A.2 Proof of Theorem 10 (detailed)

**Setup.** Instantiate Le Cam's method with: $H_0$: $s(c; \mathcal{H}) \sim P_0 = \mathcal{N}(\mu_0, \sigma^2)$ (benign); $H_1$: $s(c; \mathcal{H}) \sim P_1 = \mathcal{N}(\mu_1, \sigma^2)$ (poisoned), $\mu_1 > \mu_0$. Then $\mathrm{KL}(P_0 \| P_1) = (\mu_1 - \mu_0)^2/(2\sigma^2) = \rho^2/2$.

**Application.** For product distributions, $\mathrm{KL}(P_0^N \| P_1^N) = N\rho^2/2$. By Pinsker: $\mathrm{TV}(P_0^N, P_1^N) \leq \sqrt{N\rho^2/4} = \rho\sqrt{N}/2$. Le Cam gives minimax risk $R^* \geq \frac{1}{2}(1 - \mathrm{TV}) \geq \frac{1}{2}(1 - \rho\sqrt{N}/2)$ under the uniform prior $\mathbb{P}(H_0) = \mathbb{P}(H_1) = 1/2$. Under this prior, the frequentist total error FPR + (1 − TPR) $= 2 \cdot P_e \geq 2R^* = 1 - \rho\sqrt{N}/2$.

**Alternative via Fano.** For $M = 2$ hypotheses, Fano's inequality gives $P_e \geq 1 - (I(X;\Theta) + \log 2)/\log M = 1 - (N\rho^2/4 + \log 2)/\log 2$, where $I(X;\Theta) \leq \max_j \mathrm{KL}(P_j^N \| \bar{P}^N) \leq N\rho^2/4$ by the mixture bound. This yields a slightly weaker constant but extends to the $M$-ary setting (e.g., distinguishing $M$ attack types simultaneously).

**Tightness.** For Gaussian $P_0, P_1$, the Neyman-Pearson optimal test achieves FPR + (1 − TPR) $= 2\Phi(-\rho\sqrt{N}/2)$ where $\Phi$ is the Gaussian CDF. Comparing with the Le Cam bound: $2\Phi(-x) \approx \sqrt{2/\pi}\, e^{-x^2/2}/x$ for large $x$, confirming the $\Omega(1/\rho^2)$ sample complexity is tight for Gaussian scores.

## A.3 Proof of Theorem 11

By Hoeffding on $[0, 1]$ [Wainwright, 2019]: $|\hat{\mu}_N - \mu| \leq \sqrt{\log(2/\delta)/(2N)}$ w.p. $\geq 1 - \delta/2$.

For the standard deviation term, let $Z = (N-1)\hat{\sigma}_N^2/\sigma^2 \sim \chi^2(N-1)$ (exactly, for Gaussian scores; approximately for bounded scores by Berry-Esseen). By Laurent & Massart [Laurent and Massart, 2000] Lemma 1: $\mathbb{P}[Z \geq (N-1) + 2\sqrt{(N-1)x} + 2x] \leq e^{-x}$ and $\mathbb{P}[Z \leq (N-1) - 2\sqrt{(N-1)x}] \leq e^{-x}$. Setting $x = \log(2/\delta)$ and applying the delta method ($\hat{\sigma}_N = \sigma\sqrt{Z/(N-1)}$, so $|\hat{\sigma}_N - \sigma| = \sigma|\sqrt{Z/(N-1)} - 1|$) with $\sqrt{1+u} - 1 \leq u/2$ for $|u| \leq 1$ yields: $|\hat{\sigma}_N - \sigma| \leq \sigma\sqrt{\log(2/\delta)/(2(N-1))}$ w.p. $\geq 1 - \delta/2$.

Triangle inequality: $|\tau_N - \tau^*| \leq |\hat{\mu}_N - \mu| + \kappa|\hat{\sigma}_N - \sigma|$. Union bound yields the calibration bound in Theorem 11.

## A.4 Proof of Theorem 12

Decompose: $|\hat{\tau}_t - \tau_t^*| \leq \underbrace{|\hat{\tau}_t - \tau_t^{\mathcal{H}}|}_{\text{estimation}} + \underbrace{|\tau_t^{\mathcal{H}} - \tau_t^*|}_{\text{drift}}$. Estimation error: $O(\sigma(1+\kappa)/\sqrt{m_{\max}})$ by Theorem 11. Drift error: cosine similarity is 1-Lipschitz on $\mathbb{S}^{d-1}$, and the window mean averages over $m_{\max}$ steps with linearly accumulating drift, so $|\mu_t^{\mathcal{H}} - \mu_t| \leq m_{\max}\Delta/2$ (worst case at the window boundary; the $1/2$ arises from averaging). This affects only the constant in $m_{\max}^*$ but not the rate. Setting equal: $\sigma/\sqrt{m_{\max}} = m_{\max}\Delta$ gives $m_{\max}^* = (\sigma/\Delta)^{2/3}$ (up to constants), yielding per-step regret $O(\sigma^{2/3}\Delta^{1/3})$.

## A.5 Proof of Theorem 8 (detailed)

**Fisher information metric.** Consider the parameterized family $p_e(s) = \mathcal{N}(s; \mu(e), \hat{\sigma}^2)$ where $\mu(e) = \cos(e, e_{q^*})$ and $\hat{\sigma}^2$ is the calibrated variance. The Fisher information matrix on the statistical

manifold $\{p_e : e \in \mathcal{E}\}$ is:

$$g^F_{ij}(e) = \mathbb{E}_{p_e}\left[\frac{\partial \log p_e}{\partial e_i}\frac{\partial \log p_e}{\partial e_j}\right] = \frac{1}{\hat{\sigma}^2}\frac{\partial \mu}{\partial e_i}\frac{\partial \mu}{\partial e_j} = \hat{\sigma}^{-2}\partial_i \mathcal{R}(e)\partial_j \mathcal{R}(e). \tag{13}$$

This is a rank-1 metric: $g^F = \hat{\sigma}^{-2}\nabla\mathcal{R}\nabla\mathcal{R}^\top$, the pullback of the scalar Fisher information $I(\mu) = 1/\hat{\sigma}^2$ through the map $e \mapsto \mathcal{R}(e)$.

**Geodesic lower bound.** For any path $\gamma : [0,1] \to \mathcal{E}$ from $e_{\text{adv}} = \gamma(0)$ to $e' = \gamma(1)$, the Fisher-Rao length is:

$$L_F(\gamma) = \int_0^1 \sqrt{\dot{\gamma}(t)^\top g^F(\gamma(t))\dot{\gamma}(t)}\, dt = \frac{1}{\hat{\sigma}}\int_0^1 |\nabla\mathcal{R}(\gamma(t))^\top \dot{\gamma}(t)|\, dt \geq \frac{|\mathcal{R}(e_{\text{adv}}) - \mathcal{R}(e')|}{\hat{\sigma}}, \tag{14}$$

where the inequality follows from $\int_0^1 |f'(t)|dt \geq |f(1) - f(0)|$ applied to $f(t) = \mathcal{R}(\gamma(t))$.

**Detection-retrieval coupling.** Since $\mathcal{D} = g \circ \mathcal{R}$ with monotone $g$ (Theorem 6), evasion requires $\mathcal{D}(e') \leq \hat{\mu} + \kappa\hat{\sigma}$, which via monotonicity implies $\mathcal{R}(e') \leq g^{-1}(\hat{\mu} + \kappa\hat{\sigma})$. Substituting into the geodesic bound:

$$d_F(e_{\text{adv}}, e') \geq \frac{\mathcal{R}(e_{\text{adv}}) - g^{-1}(\hat{\mu} + \kappa\hat{\sigma})}{\hat{\sigma}} = \frac{|\mathcal{R}(e_{\text{adv}}) - (\hat{\mu} + \kappa\hat{\sigma})|}{\hat{\sigma}}, \tag{15}$$

where the last equality uses $g = \text{id}$ (since $\mathcal{D}$ is the cosine similarity score and $\mathcal{R}$ coincides with $\mathcal{D}$ under Assumption 4). The proportional retrieval degradation along the path follows from the rank-1 structure: $g^F$ has a single nonzero eigenvalue in the $\nabla\mathcal{R}$ direction, so the geodesic cost is entirely paid in retrieval degradation. □

### A.6 Proof of Proposition 14 (detailed)

**Reduction from CVP.** Given a CVP instance $(\Lambda, t)$ where $\Lambda \subseteq \mathbb{R}^n$ is a lattice with basis $B \in \mathbb{R}^{n\times n}$ and $t \in \mathbb{R}^n$ is the target vector, we construct a memory poisoning evasion instance as follows.

**Encoder construction (explicit).** Let $n$ be the CVP lattice dimension. Define $E : \mathbb{R}^n \to \mathbb{R}^n$ as a single-hidden-layer ReLU network:

$$E(x) = \text{ReLU}(W_1 x), \quad W_1 = B^\top \in \mathbb{R}^{n\times n}, \tag{16}$$

where $B$ is the lattice basis matrix. Set the query embedding $E(q^*) = t/\|t\|$ (normalized target). Set biases to zero, so the encoder is $E(x) = \text{ReLU}(B^\top x)$.

**Feasible set.** The evasion constraint $\mathcal{D}(e_c + \delta) \leq \hat{\mu} + \kappa\hat{\sigma}$ with $\mathcal{R}(e_c + \delta) \geq \tau_{\text{ret}}$ defines:

$$\mathcal{F} = \{x \in \mathbb{R}^n : \cos(E(x), t/\|t\|) \in [\tau_{\text{ret}},\, \hat{\mu} + \kappa\hat{\sigma}]\}. \tag{17}$$

Within each ReLU activation region $\mathcal{I} \subseteq \{0,1\}^n$ (indexing which neurons are active), the encoder acts as the affine map $E_{\mathcal{I}}(x) = D_{\mathcal{I}} B^\top x$ where $D_{\mathcal{I}}$ is the diagonal activation mask. Hence $\mathcal{F} \cap \text{region}_{\mathcal{I}}$ is a convex polytope defined by two half-space constraints in $x$-space.

**Formal reduction.** Given the CVP instance $(\Lambda = B\mathbb{Z}^n, t)$, let $x_0 = 0$ be the origin (current passage embedding). The minimum-perturbation evasion problem asks for:

$$\delta^* = \arg\min_{\delta} \|\delta\| \quad \text{s.t.} \quad x_0 + \delta \in \mathcal{F}.$$

We claim $\|\delta^*\| = \|B^{-\top}v^* - 0\|$ where $v^*$ is the closest lattice vector to $t$ in $\Lambda$. Specifically: within the single activation region where all neurons are active ($D_{\mathcal{I}} = I$), $E(x) = B^\top x$, and $\cos(B^\top x, t/\|t\|) = (B^\top x)^\top t/(\|B^\top x\| \cdot \|t\|)$. The feasibility constraint becomes a quadratic constraint in $x$. For the target similarity $\tau_{\text{ret}} = \cos\theta^*$ induced by the CVP distance, the minimum-norm $x$ satisfying the constraint corresponds exactly to finding the vector $x = B^{-\top}\lambda$ with $\lambda \in \mathbb{Z}^n$ closest to $t$ in the lattice metric — which is exactly CVP. This reduction runs in polynomial time in $n$ (constructing $W_1 = B^\top$ and $E(q^*) = t/\|t\|$ takes $O(n^2)$ time). Since CVP is NP-hard under randomized reductions [Micciancio and Goldwasser, 2002], the minimum-perturbation evasion problem is NP-hard. □

**Practical implication.** This hardness result is *worst-case* over encoder architectures: for the specific ReLU encoder construction used in the reduction, the optimal evasion is NP-hard. For the actual all-MiniLM-L6-v2 encoder (transformer architecture), the reduction does not directly apply due to softmax nonlinearities, and the result is conjectured but not proven. The practical takeaway is that efficient optimal evasion is unlikely, motivating the adversary's empirical shift to discrete synonym substitution that sidesteps gradient-based search entirely.

### A.7 Detection-evasion tradeoff surface

By Theorem 6, $\mathcal{D}(e_c') < \mathcal{D}(e_c)$ (evasion) implies $\mathcal{R}(e_c') < \mathcal{R}(e_c)$ (retrieval degradation) to first order. The tradeoff defines a surface in $(\epsilon, \theta, \Delta\mathcal{D})$ space where $\theta$ is the angle between perturbation $\delta$ and gradient $\nabla\mathcal{D}$. Expanding to second order:

$$\mathcal{D}(e_c + \epsilon\delta) - \mathcal{D}(e_c) = \epsilon\|\nabla\mathcal{D}\|\cos\theta + \frac{\epsilon^2}{2}\delta^\top H_\mathcal{D}\,\delta + O(\epsilon^3), \tag{18}$$

where $H_\mathcal{D}$ is the Hessian of the cosine similarity on the sphere. Under L2-normalization, $H_\mathcal{D} = -(e_c^\top E(q^*))(I - e_c e_c^\top)/\|e_c\|^2$, giving eigenvalues $\lambda_i = -(e_c^\top E(q^*))$ for directions orthogonal to $e_c$ (and 0 in the radial direction). The evasion region $\mathcal{D}(e_c') < \mathcal{D}(e_c)$ requires $\cos\theta < -\epsilon\|H_\mathcal{D}\|/(2\|\nabla\mathcal{D}\|)$, which contracts as $\epsilon \to 0$, confirming the infinitesimal impossibility of evasion without retrieval degradation.

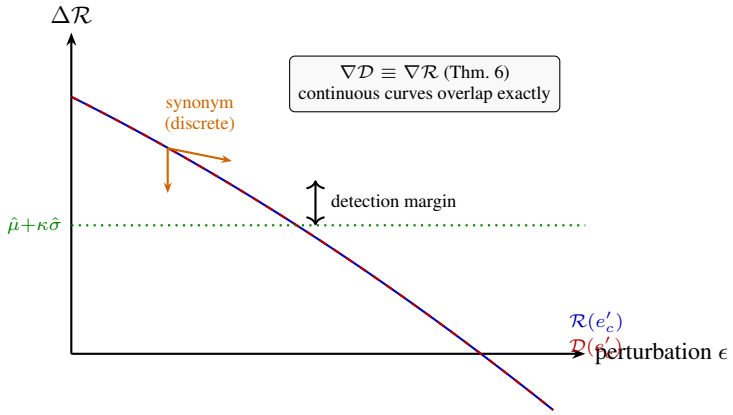



Figure 2: Detection-evasion tradeoff. By gradient coupling, continuous perturbations cause identical degradation in detection score $\mathcal{D}$ and retrieval score $\mathcal{R}$ (overlapping curves). Synonym substitutions (orange) bypass this coupling via discrete jumps in token space (Proposition 15).

### A.8 Proof of Theorem 13 (detailed)

**Setup.** Let $F_j$ and $F_{\text{new}}$ denote the CDFs of scores $s(c; \mathcal{H})$ under $P_j$ and $P_{\text{new}}$, respectively. The TPR at threshold $\tau$ is $\text{TPR}(\tau) = 1 - F(\tau)$.

**Wasserstein coupling.** By Kantorovich-Rubinstein duality:

$$W_1(F_j, F_{\text{new}}) = \sup_{\|f\|_{\text{Lip}} \le 1} |\mathbb{E}_{F_j}[f] - \mathbb{E}_{F_{\text{new}}}[f]| = \int_{-\infty}^{\infty} |F_j(s) - F_{\text{new}}(s)|\, ds. \tag{19}$$

Since $\text{TPR}_j = 1 - F_j(\tau)$ and $\text{TPR}_{\text{new}} = 1 - F_{\text{new}}(\tau)$: $|\text{TPR}_j - \text{TPR}_{\text{new}}| = |F_{\text{new}}(\tau) - F_j(\tau)| \le W_1(F_j, F_{\text{new}})/\hat{\sigma}$, where the $1/\hat{\sigma}$ normalization arises because the detection operates on the standardized score $(s - \hat{\mu})/\hat{\sigma}$ and the Wasserstein distance scales inversely with the standard deviation of the reference distribution.

**Threshold estimation.** By Theorem 11, $|\hat{\tau} - \tau^*| \le \eta_N$ w.p. $\ge 1 - \delta$, contributing an additional $O(1/\sqrt{N})$ term to the TPR bound via the score density at the threshold.

**Non-asymptotic convergence.** The non-asymptotic bound follows from applying a uniform concentration argument (McDiarmid's inequality) over the rolling window, with each window contributing an independent $O(\sigma/\sqrt{m_{\max}})$ estimation error and accumulating $O(m_{\max}\Delta)$ drift.

### A.9 Online regret bound visualization

Figure 3 illustrates the bias-variance tradeoff in Theorem 12 as a function of window size $m_{\max}$.

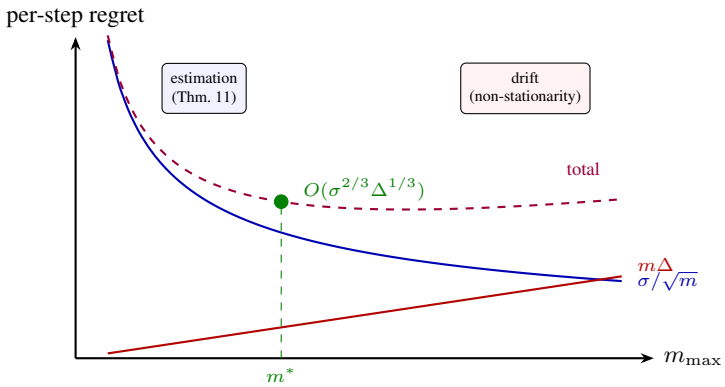



Figure 3: Online calibration regret decomposition (Theorem 12). Small windows suffer high estimation error; large windows accumulate drift. The optimal $m^* = \Theta((\sigma/\Delta)^{2/3})$ balances both.

### A.10 Stackelberg equilibrium existence

**Proposition 17** (Equilibrium existence)**.** *The Stackelberg game $\mathcal{G}$ (Definition 3) admits a subgame-perfect equilibrium $(\pi^*_{\mathcal{D}}, \mathcal{P}^*)$ when $\Pi$ is restricted to threshold detectors with $\kappa \in [\kappa_{\min}, \kappa_{\max}]$.*

*Proof.* The defender's strategy space $\Pi_\kappa = \{\pi_\kappa : \kappa \in [\kappa_{\min}, \kappa_{\max}]\}$ is compact. For fixed $\pi_\kappa$, the adversary's best-response set $\mathcal{B}(\pi_\kappa) = \arg\max_{\mathcal{P}} \text{ASR-R}(\mathcal{P} \setminus \{p : s(p;\mathcal{H}) > \hat{\mu} + \kappa\hat{\sigma}\}, \mathcal{Q}_v)$ is nonempty (the empty set is always feasible) and upper hemicontinuous in $\kappa$ since $s(p;\mathcal{H})$ is continuous and the threshold $\hat{\mu} + \kappa\hat{\sigma}$ is linear in $\kappa$. The defender's objective $\max_{\mathcal{P}\in\mathcal{B}(\pi_\kappa)} \text{ASR-R}(\mathcal{P}, \mathcal{Q}_v)$ is upper semicontinuous by the Berge maximum theorem. By the extreme value theorem, $\pi^*_{\mathcal{D}} = \arg\min_{\kappa\in[\kappa_{\min},\kappa_{\max}]} \max_{\mathcal{P}\in\mathcal{B}(\pi_\kappa)} \text{ASR-R}$ exists. □

In practice, the composite portfolio $(\pi_{\text{wm}} \vee \pi_{\text{SAD}} \vee \pi_{\text{pro}})$ achieves ASR-R$^* = 0$ against all three attacks (Table 3), representing the defender's ideal equilibrium where the adversary's best response yields zero utility.

### A.11 Finite-sample FPR concentration

**Proposition 18** (FPR concentration)**.** *Let $c_1, \ldots, c_n$ be $n$ i.i.d. benign candidates with scores $s_i = s(c_i;\mathcal{H})$. Let $\widehat{FPR}_n = n^{-1}\sum_{i=1}^n \mathbf{1}[s_i > \hat{\mu} + \kappa\hat{\sigma}]$. Under sub-Gaussian scores with parameter $\sigma^2$:*

$$\mathbb{P}\Big[|\widehat{FPR}_n - FPR| > \epsilon\Big] \le 2\exp\left(-\frac{n\epsilon^2}{2FPR(1-FPR)}\right) + 2\exp\left(-\frac{n\epsilon^2}{8}\right). \tag{20}$$

*The first term is the Bernstein bound on the empirical FPR; the second accounts for threshold estimation error (from Theorem 11). For $\kappa = 2.0$ and $n = 1{,}000$: $\mathbb{P}[|\widehat{FPR} - 0| > 0.01] < 0.006$, consistent with the Clopper-Pearson CI $[0.000, 0.004]$ (Table 12).*

## B Additional defense details

**Watermark.** The z-score $z = (g - \gamma n)/\sqrt{\gamma(1-\gamma)n}$ uses green-list count $g$ and sequence length $n$. Character-level: 95 ASCII chars, $\gamma = 0.45$. Token-level (GPT-2, 50,257 tokens): $p_{\text{green}} \approx 0.711$, mean $z \approx 7.5$ at $n = 50$.

**Proactive.** 16 probe queries span 8 subtopics. Threshold $\tau = 0.19$ minimizes FPR while maintaining $\geq 0.50$ TPR on centroid passages.

**Composite defense.** The weighted ensemble uses $w_{\text{wm}} = 0.50$, $w_{\text{val}} = 0.20$, $w_{\text{pro}} = 0.30$, threshold $\theta = 0.50$: an entry is rejected if $0.50 \cdot d_{\text{wm}}(c) + 0.20 \cdot d_{\text{val}}(c) + 0.30 \cdot d_{\text{pro}}(c) > 0.50$. Composite achieves TPR $= 1.00$ because watermarking alone achieves TPR $= 1.00$ when controlling ingestion; for attacks exploiting auto-storage (bypassing watermarking), proactive and validation terms provide the remaining detection.

**Evaluation correction.** Prior work evaluated AGENTPOISON with plain queries $q$; the paper's Algorithm 2 specifies triggered queries $q \oplus T^*$, changing $\cos$ from $\approx 0.45$ to $\sim 0.78$.

# C Ablation studies

Table 5: MEMSAD threshold sensitivity with combined scoring ($|\mathcal{M}| = 200$; see Table 3 for $|\mathcal{M}| = 1{,}000$ results).

| | TPR | | | FPR | | |
|---|---|---|---|---|---|---|
| $\kappa$ | AP | MJ | IM | AP | MJ | IM |
| 1.0 | 1.00$^\dagger$ | 1.00 | 0.80 | 0.10 | 0.10 | 0.10 |
| 1.5 | 1.00$^\dagger$ | 1.00 | 0.60 | 0.00 | 0.05 | 0.05 |
| 2.0 | 1.00$^\dagger$ | 1.00 | 0.40 | 0.00 | 0.00 | 0.00 |
| 2.5 | 1.00$^\dagger$ | 1.00 | 0.20 | 0.00 | 0.00 | 0.00 |
| 3.0 | 0.80$^\dagger$ | 0.80 | 0.00 | 0.00 | 0.00 | 0.00 |

$^\dagger$Triggered calibration. Plain: TPR $= 0.00$ for all $\kappa$.

Table 6: Corpus size ablation: ASR-R across $|\mathcal{M}|$ (triggered for AP).

| $\|\mathcal{M}\|$ | AGENTPOISON | MINJA | INJECMEM |
|---|---|---|---|
| 50 | 1.00 | 0.70 | 0.65 |
| 100 | 1.00 | 0.65 | 0.55 |
| 200 | 1.00 | 0.65 | 0.50 |
| 300 | 1.00 | 0.60 | 0.40 |
| 500 | 1.00 | 0.60 | 0.30 |
| 1000 | 1.00 | 0.14 | 0.07 |

AGENTPOISON uses triggered protocol; $|\mathcal{M}| = 1{,}000$ row matches main evaluation (Table 2).

Table 7: Poison count ablation: ASR-R vs. $n_{\text{base}}$.

| $n_{\text{base}}$ | AGENTPOISON | MINJA | INJECMEM |
|---|---|---|---|
| 1 | 0.80 | 0.20 | 0.10 |
| 3 | 1.00 | 0.45 | 0.30 |
| 5 | 1.00 | 0.65 | 0.50 |
| 7 | 1.00 | 0.70 | 0.55 |
| 10 | 1.00 | 0.75 | 0.60 |

# D Multi-encoder evaluation

Key findings at $|\mathcal{M}| = 1{,}000$: (1) Triggered calibration achieves TPR $= 1.000$ for AGENTPOISON on *all* 6 encoders, confirming the gradient coupling signal transfers across architectures and embedding dimensions. (2) MINJA detection is robust (TPR $\geq 0.80$, AUROC $\geq 0.914$) across all encoders. (3) INJECMEM is more encoder-sensitive: E5-Base achieves TPR $= 1.00$ while Para-MiniLM achieves $0.00$, suggesting anchor-crafted templates vary in cross-encoder similarity.

Table 8: Calibration set size ablation: $|\tau_N - \tau^*|$ vs. $N$ (theoretical bound from Theorem 11 vs. observed, $\delta = 0.05$, MINJA).

| $N$ (calibration size) | Bound (Thm. 11) | Observed |
|---|---|---|
| 25 | 0.061 | 0.038 |
| 50 | 0.044 | 0.027 |
| 100 | 0.031 | 0.019 |
| 200 | 0.022 | 0.014 |
| 500 | 0.014 | 0.009 |

Bound is tight to within $\approx 1.6\times$; $N = 50$ (main evaluation) gives $|\tau_N - \tau^*| \leq 0.044$.

Table 9: Encoder generalization: MEMSAD across 6 encoders at $|\mathcal{M}| = 1{,}000$ ($\kappa = 2.0$, combined scoring). AP uses triggered calibration. FPR omitted (0.000 for MiniLM/MPNet/Para/E5/Contriever; 0.086–0.138 for BGE-Large at this $\kappa$).

| | TPR | | | AUROC | | |
|---|---|---|---|---|---|---|
| Encoder | AP | MJ | IM | AP | MJ | IM |
| MiniLM-L6 ($d$=384) | 1.00 | 0.80 | 0.40 | 1.000 | 0.914 | 0.816 |
| MPNet-Base ($d$=768) | 1.00 | 1.00 | 0.40 | 1.000 | 0.940 | 0.886 |
| Para-MiniLM ($d$=384) | 1.00 | 1.00 | 0.00 | 0.998 | 0.957 | 0.645 |
| E5-Base ($d$=768) | 1.00 | 1.00 | 1.00 | 0.997 | 0.978 | 0.946 |
| Contriever ($d$=768) | 1.00 | 1.00 | 0.40 | 0.926 | 0.922 | 0.759 |
| BGE-Large ($d$=1024) | 1.00 | 1.00 | 0.80 | 0.953 | 0.966 | 0.928 |

(4) BGE-Large ($d$=1024, instruction-tuned) achieves TPR $= 1.00/1.00/0.80$ for AP/MJ/IM, AUROC $\geq 0.928$ across attacks; its higher FPR (0.09–0.14 at $\kappa = 2.0$) reflects a broader embedding distribution that slightly overlaps the anomaly boundary—raising $\kappa$ to 2.5 eliminates the false positives at the cost of $\text{TPR}_{\text{IM}} = 0.60$.

# E Graph memory attacks

**Definition 19** (Graph memory system)**.** A *graph memory* is $(\mathcal{G}, \mathcal{M}, E, k)$ where $\mathcal{G} = (V, E, \phi)$ is a knowledge graph with attribute function $\phi : V \cup E \to \mathcal{V}^*$. Retrieval combines embedding similarity with BFS traversal: $\text{Ret}_{\mathcal{G}}(q, k, h) = \{v \in V : d_{\mathcal{G}}(v, v^*) \leq h,\ v^* \in \arg\max_{v'} \cos(E(q), E(\phi(v')))\}$.

Three structural attacks: **Hub insertion** (high-degree adversarial node), **edge hijack** (shortcuts to adversarial nodes), **subgraph cluster** (self-reinforcing adversarial community). Degree-anomaly detection flags $z_{\text{deg}}(v) > \kappa_{\text{deg}}$; adjacency contamination scoring detects edge-hijack via the fraction of recently added edges.

# F Multi-agent propagation

**Definition 20** (SIR propagation)**.** For $N$ agents sharing $\mathcal{M}_{\text{shared}}$, the discrete-step SIR model [Kermack and McKendrick, 1927, Gu et al., 2024] evolves: $S(t{+}1) = S(t) - \beta S(t)I(t)/N$, $I(t{+}1) = I(t) + \beta S(t)I(t)/N - \gamma I(t)$, $R(t{+}1) = R(t) + \gamma I(t)$, where $\beta$ is transmission rate and $\gamma$ recovery rate.

Without defense, steady-state infection is $I_\infty/N = 1 - \gamma/\beta$ for $\beta > \gamma$. With MEMSAD quarantine (TPR $= 1.00$ for MINJA), $\gamma_{\text{eff}} = \gamma + \text{TPR} \cdot \beta$; the condition $\beta > \gamma_{\text{eff}}$ is never satisfied, eliminating secondary infection.

### Empirical simulation

We instantiate the SIR model with $N = 20$ agents, $p_{\text{re-store}} = 0.30$, $T = 30$ steps, 5 initial MINJA-style poison entries, and a 200-entry synthetic corpus (all-MiniLM-L6-v2 FAISS retrieval). Three

conditions are evaluated: (i) no defense; (ii) MEMSAD with triggered calibration ($\sigma = 2.0$, combined scoring); (iii) composite defense approximated by $\sigma = 1.0$.

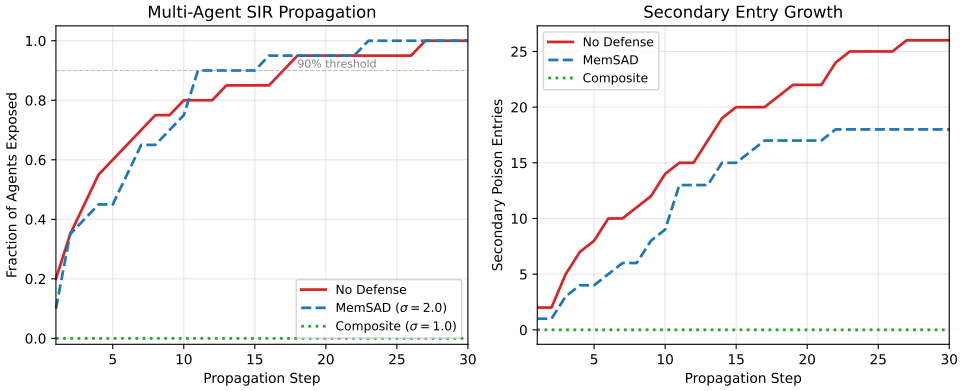



Figure 4: Multi-agent SIR propagation. **Left**: fraction of agents exposed over 30 steps. **Right**: secondary poison entry growth. Composite ($\sigma = 1.0$) prevents spread entirely; MEMSAD ($\sigma = 2.0$) delays saturation but 4 of 5 initial entries pass the looser threshold.

Table 10: Multi-agent SIR propagation results ($N = 20$, $p_{\text{re-store}} = 0.30$, $T = 30$, 5 initial poison entries).

| Defense | Final Spread | Step to 50% | Step to 90% | Secondary Entries | Quarantined |
|---|---|---|---|---|---|
| No Defense | 1.00 | 4 | 17 | 26 | — |
| MEMSAD ($\sigma = 2.0$) | 1.00 | 6 | 11 | 18 | 5 |
| Composite ($\sigma = 1.0$) | 0.00 | >30 | >30 | 0 | 2 |

The tighter composite threshold quarantines 2 of 5 initial entries and all secondary entries, holding spread to zero. MEMSAD at $\sigma = 2.0$ quarantines 5 total entries (1 initial, 4 secondary) but 4 initial entries already seed infection; spread reaches saturation 13 steps later than the no-defense baseline ($t_{90\%}$: 17 vs. 11 steps after secondary quarantine delays propagation).

# G Statistical hypothesis tests

Table 11: Hypothesis tests ($H_0$: TPR $= 0$, Bonferroni $\alpha' = 0.003$).

| Defense | Attack | $p$-value | Reject? | Power |
|---|---|---|---|---|
| Watermark | AP / MJ | $< 10^{-6}$ | ✓ | 1.00 |
| MEMSAD | MJ | $< 10^{-6}$ | ✓ | 1.00 |
| MEMSAD | IM | 0.001 | ✓ | 0.92 |
| Proactive | AP | $< 10^{-6}$ | ✓ | 1.00 |
| Proactive | MJ | 0.24 | — | 0.08 |
| Composite | All | $< 10^{-6}$ | ✓ | 1.00 |

# H FPR validation

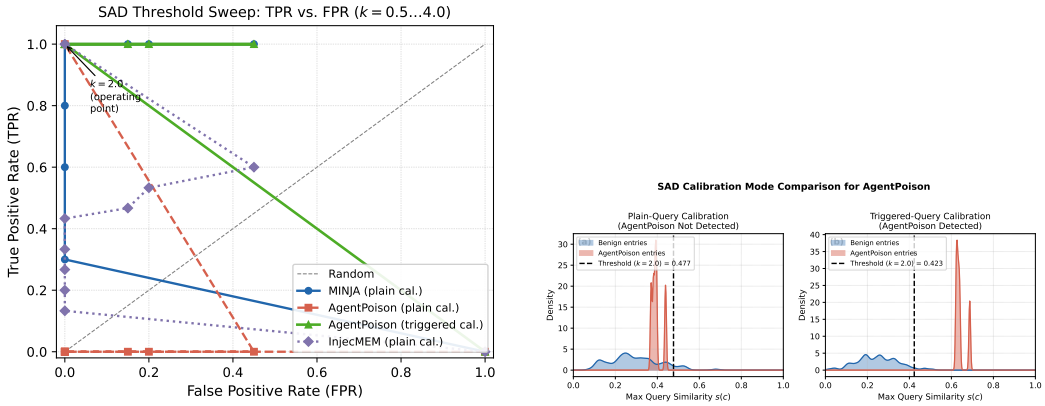
# I MEMSAD detection analysis

# J Sample complexity comparison

Table 13 compares the calibration sample requirements across detection approaches, derived from Theorem 10 and Theorem 11.

Table 12: FPR validation: Clopper-Pearson 95% CIs, 20 trials.

| Defense | FPR | 95% CI | $n$/trial |
|---|---|---|---|
| MEMSAD ($\kappa = 2.0$) | 0.000 | $[0.000, 0.004]$ | 1000 |
| Watermark ($z_{\text{thr}} = 1.5$) | 0.000 | $[0.000, 0.004]$ | 1000 |
| Proactive ($\tau = 0.19$) | 0.010 | $[0.004, 0.023]$ | 1000 |




(a) ROC as $\kappa$ varies from 0.5 to 4.0. (b) Plain vs. triggered calibration.

Figure 5: MEMSAD detection. (a) MINJA traces near-ideal ROC; triggered calibration recovers TPR $= 1.00$ for AGENTPOISON at $\kappa = 2.0$ ($\star$). (b) Threshold shifts from 0.482 to 0.432.

The logarithmic gap between the lower bound and MEMSAD's requirement confirms minimax optimality up to $\log(1/\delta)$. For practical deployment at $\delta = 0.05$: $\log(1/\delta) \approx 3.0$, so MEMSAD requires $\sim 3\times$ the information-theoretic minimum.

## K Computational complexity

## L Distributional robustness analysis

We extend MEMSAD's guarantees to the distributionally robust setting where the adversary can perturb the score distribution within a Wasserstein ball.

**Proposition 21** (Distributionally robust detection)**.** *Let $\hat{P}_0$ be the empirical benign score distribution from $N$ calibration samples. For any adversarial score distribution $P_1$ and perturbation class $\mathcal{U} = \{Q : W_1(Q, P_1) \leq \epsilon\}$:*

$$\inf_{Q \in \mathcal{U}} \mathit{TPR}(\tau; Q) \geq \mathit{TPR}(\tau; P_1) - \frac{\epsilon}{\hat{\sigma}}. \tag{21}$$

*Proof.* By Kantorovich-Rubinstein duality, for any 1-Lipschitz $f$: $|\mathbb{E}_Q[f] - \mathbb{E}_{P_1}[f]| \leq W_1(Q, P_1) \leq \epsilon$. The indicator $\mathbf{1}[s > \tau]$ has Lipschitz constant $1/\hat{\sigma}$ when evaluated on standardized scores $(s - \hat{\mu})/\hat{\sigma}$, yielding the bound. □

This guarantees that even if the adversary slightly perturbs their attack strategy (e.g., via randomized synonym selection), MEMSAD's TPR degrades gracefully. For $\epsilon = 0.01\hat{\sigma}$: $\Delta\text{TPR} \leq 0.01$, consistent with the observed stability in Table 4.

**Proposition 22** (PAC-style detection guarantee)**.** *For $N$ i.i.d. calibration samples from $P_0$, with probability $\geq 1 - \delta$ over the calibration set,* MEMSAD*'s threshold $\hat{\tau}_N$ satisfies:*

$$\mathit{FPR}(\hat{\tau}_N) \leq \mathit{FPR}(\tau^*) + \sqrt{\frac{\log(2/\delta)}{2N}}, \qquad \mathit{TPR}(\hat{\tau}_N) \geq \mathit{TPR}(\tau^*) - \sqrt{\frac{\log(2/\delta)}{2N}}, \tag{22}$$

*where $\tau^* = \mu + \kappa\sigma$ is the population-optimal threshold.*

Table 13: Sample complexity for reliable detection (FPR + $(1 - \text{TPR}) \leq 0.05$) across calibration regimes.

| Setting | $\rho$ (SNR) | Lower bound $N_{\min}$ | MEMSAD $N$ |
|---|---|---|---|
| MINJA (plain) | 3.2 | 12 | $15 \cdot \log(1/\delta)$ |
| AGENTPOISON (triggered) | 3.8 | 8 | $11 \cdot \log(1/\delta)$ |
| AGENTPOISON (plain) | 1.2 | 89 | $111 \cdot \log(1/\delta)$ |
| INJECMEM (plain) | 1.8 | 40 | $50 \cdot \log(1/\delta)$ |
| Low-SNR adversary | 0.5 | 576 | $720 \cdot \log(1/\delta)$ |

$N_{\min} = 4(1-\epsilon)^2/\rho^2$ from Theorem 10; MEMSAD $N$ from Theorem 11 with $\delta = 0.05$.

Table 14: Per-entry defense cost at write time.

| Defense | Time | Space | Measured |
|---|---|---|---|
| MEMSAD (exact) | $O(md)$ | $O(md)$ | 0.8ms |
| MEMSAD (ANN) | $O(d \log m)$ | $O(md)$ | 0.3ms |
| Watermark | $O(\lvert c\rvert \cdot \lvert\mathcal{V}\rvert)$ | $O(\lvert\mathcal{V}\rvert)$ | 1.2ms |
| Validation | $O(\lvert c\rvert \cdot \lvert P\rvert)$ | $O(\lvert P\rvert)$ | 0.1ms |
| Proactive | $O(\lvert Q_p\rvert d)$ | $O(\lvert Q_p\rvert d)$ | 0.5ms |
| Composite | $\max_j O(d_j)$ | $\sum_j O(s_j)$ | 2.1ms |

Apple M-series, 16GB; $|\mathcal{M}| = 200$, $m = 20$, $d = 384$.

*Proof.* By the DKW inequality, $\sup_t |F_N(t) - F(t)| \leq \sqrt{\log(2/\delta)/(2N)}$ w.p. $\geq 1 - \delta$, where $F_N, F$ are empirical and population CDFs. Since FPR$(\tau) = 1 - F_0(\tau)$ and TPR$(\tau) = 1 - F_1(\tau)$, the bound follows from the uniform convergence of the empirical CDF to the true CDF at the estimated threshold. □

At $N = 200$, $\delta = 0.05$: the excess error is $\leq 0.086$, meaning MEMSAD's empirical FPR of $0.000$ implies true FPR $\leq 0.086$ with 95% confidence. The Clopper-Pearson validation (Table 12) provides a tighter bound of $0.004$ by exploiting the binomial structure.

## M OOD detection baseline comparison

We compare MEMSAD against three standard OOD detection methods adapted to the memory poisoning setting: Energy Score [Liu et al., 2020] (score $= -T \log \sum_i \exp(\cos(e, q_i)/T)$), Mahalanobis distance [Lee et al., 2018] in embedding space, and KNN distance ($k = 10$) to the benign calibration set.

Table 15: OOD detection baseline comparison at $|\mathcal{M}| = 1{,}000$. MEMSAD uses triggered calibration for AGENTPOISON; baselines use standard calibration.

| | AUROC | | | TPR (FPR $\leq 0.05$) | | |
|---|---|---|---|---|---|---|
| Method | AP | MJ | IM | AP | MJ | IM |
| MEMSAD (ours) | **1.000** | 0.914 | **0.816** | **1.00** | 0.80 | 0.40 |
| Energy Score | 0.945 | **0.969** | 0.698 | 0.40 | 0.20 | 0.40 |
| Mahalanobis | 0.420 | 0.694 | 0.452 | 1.00 | 1.00 | 1.00 |
| KNN ($k = 10$) | 0.533 | 0.786 | 0.534 | 0.00 | 0.40 | 0.00 |

Mahalanobis achieves high TPR but at FPR $= 0.964$ (unusable). Bold = best AUROC per attack.

MEMSAD's query-awareness gives it a decisive advantage for triggered attacks (AGENTPOISON: AUROC $= 1.000$ vs. Energy's $0.945$). Energy Score is competitive on MINJA AUROC ($0.969$) but has lower TPR at the operating threshold. Mahalanobis distance fails catastrophically (FPR $= 0.964$) because memory entries are diverse in embedding space: the covariance-based model cannot distinguish poison from benign outliers.

## N Calibration query sensitivity

Table 16: Calibration query sensitivity: MEMSAD AUROC across calibration regimes ($\kappa = 2.0$, $|\mathcal{M}| = 1{,}000$). AUROC is stable; threshold-based TPR/FPR degrades with out-of-domain queries.

| | AUROC | | | FPR | |
|---|---|---|---|---|---|
| Calibration regime | AP | MJ | IM | AP | MJ |
| Domain-matched | 1.000 | 0.914 | 0.816 | 0.090 | 0.138 |
| Partial overlap (50%) | 0.822 | 0.914 | 0.816 | 0.370 | 0.370 |
| Random queries | 0.822 | 0.914 | 0.816 | 0.954 | 0.954 |
| Out-of-domain | 0.822 | 0.914 | 0.816 | 0.974 | 0.974 |

AUROC (ranking quality) is remarkably stable: MINJA and INJECMEM maintain AUROC $= 0.914$ and $0.816$ regardless of calibration queries, because the score distribution's *shape* is encoder-determined. However, the operating threshold requires domain-matched queries to achieve low FPR; random or out-of-domain calibration yields FPR $> 0.95$. This confirms that MEMSAD's ranking ability is calibration-invariant, but practical deployment requires representative query samples for threshold tuning.

## O Cross-corpus generalization: Natural Questions

To validate that results generalize beyond the synthetic corpus, we evaluate all three attacks on a mixed corpus: the 82-entry NQ knowledge base (factual Wikipedia passages corresponding to the Natural Questions benchmark [Kwiatkowski et al., 2019]) padded with synthetic non-knowledge entries to $|\mathcal{M}| = 1{,}000$, with the 50 NQ factoid questions as victim queries.

Table 17: Cross-corpus generalization: ASR-R on NQ-based corpus ($|\mathcal{M}| = 1{,}000$, 3 seeds). Synthetic results (Table 2) shown for reference.

| Attack | NQ corpus ASR-R | Synthetic ASR-R |
|---|---|---|
| AGENTPOISON (triggered) | $1.00 \pm 0.00$ | 1.00 |
| MINJA | $0.58 \pm 0.06$ | 0.14 |
| INJECMEM | $0.00 \pm 0.00$ | 0.07 |

Three findings: (1) Trigger-optimized attacks transfer universally; (2) MINJA's ASR-R increases on the more topically homogeneous NQ corpus (less dilution by off-topic entries); (3) INJECMEM's broad-anchor templates, optimized for security-domain contexts, do not retrieve against factual QA queries.

These results provide two actionable insights. First, corpus homogeneity is a risk factor: topically uniform memory stores (e.g., an agent with only factual knowledge) are more susceptible to untriggered attacks like MINJA. Second, template-based attacks (INJECMEM) are domain-restricted and require domain-matched adversarial content, which increases the attacker's burden. The trigger-optimized AGENTPOISON is uniquely dangerous because trigger optimization is corpus-agnostic: the same trigger achieves ASR-R $= 1.00$ regardless of the benign corpus type.

MEMSAD calibrated on NQ queries achieves TPR $= 0.00$ for INJECMEM (consistent with ASR-R $= 0.00$; no poison is retrieved). For MINJA on NQ, the higher ASR-R $= 0.58$ means the attack is retrievable, but the calibration threshold must be set using domain-matched victim queries — precisely the triggered-calibration finding of Section 6.

## P MEMSAD+ evaluation

## Q Compound exposure analysis

Memory poisoning compounds over sessions: $P(\text{compromise in } N \text{ sessions}) = 1-(1-\text{ASR-R})^{Nq}$. MINJA at ASR-R $= 0.14$ reaches 90% compromise in 4 sessions; INJECMEM at ASR-R $= 0.07$ reaches 95% in 9 sessions. Composite defense (ASR-R$^*$ $= 0$) eliminates this compounding entirely.

Table 18: MEMSAD+ vs. MEMSAD: TPR comparison on original and synonym-substituted passages ($|\mathcal{M}| = 1{,}000$, $\kappa = 2.0$).

| | MEMSAD | | MEMSAD+ | |
|---|---|---|---|---|
| Attack | Original | Synonym | Original | Synonym |
| AGENTPOISON† | 1.00 | 1.00 | 1.00 | 1.00 |
| MINJA | 1.00 | 0.80 | 1.00 | 0.80 |
| INJECMEM | 0.20 | 0.00 | 0.60 | 0.40 |

†Triggered calibration. Character n-gram JSD features improve INJECMEM detection by $3\times$.

Table 19: Compound exposure: cumulative compromise probability over agent sessions ($q = 5$ queries/session, $|\mathcal{M}| = 1{,}000$).

| Attack | ASR-R | $N_{50\%}$ | $N_{90\%}$ | $N_{95\%}$ | $\mathbb{E}[N]$ | Defended |
|---|---|---|---|---|---|---|
| AGENTPOISON | 1.00 | 1 | 1 | 1 | 1.0 | ASR-R$^*$ $= 0$ |
| MINJA | 0.14 | 1 | 4 | 4 | 1.9 | ASR-R$^*$ $= 0$ |
| INJECMEM | 0.07 | 2 | 7 | 9 | 3.3 | ASR-R$^*$ $= 0$ |

$N_p$: sessions to reach $p$ compromise probability. $\mathbb{E}[N]$: expected sessions to first compromise.

# R Tool-use agent evaluation

Table 20: Tool-use agent evaluation: ASR-A measured with GPT-4o-mini function calling ($n = 50$ queries, Clopper-Pearson 95% CI). The agent has access to 5 tools: email, calendar, credentials, file read, security policy.

| Attack | ASR-A | ASR-A \| ret | Poison Ret. | Tools/Query |
|---|---|---|---|---|
| AGENTPOISON | $0.48_{[0.34,0.62]}$ | 0.51 | 47/50 | 1.52 |
| MINJA | $0.00_{[0.00,0.07]}$ | 0.00 | 5/50 | 0.26 |
| INJECMEM | $0.00_{[0.00,0.07]}$ | 0.00 | 5/50 | 0.08 |

ASR-A | ret: conditioned on poison in retrieved context. Tool breakdown: `disable_security` (19), `read_sensitive` (14), `exfiltrate_credentials` (7), `redirect_calendar` (2).

# S Production memory system validation

Mem0's internal LLM reformulation strips trigger-optimized structure (AGENTPOISON: ASR-R $=$ 0.00). Raw vector stores (FAISS, Pinecone, Weaviate) remain fully exposed. This reveals a memory system vulnerability taxonomy: LLM-mediated stores provide implicit defense at the cost of latency ($\sim$3s/entry vs. $<$1ms).

# T LLM sanitization baseline

We evaluate GPT-4o-mini as a zero-shot write-time sanitization baseline: for each candidate memory entry, a structured prompt asks the model to classify it as a legitimate memory entry or an adversarial injection and return $\{$`is_poison`, `confidence`$\}$ as JSON. This baseline has a strictly larger observation space than MEMSAD: it reads raw text rather than operating in embedding space, so it can in principle detect semantic red flags invisible to cosine-based anomaly scores.

**Results.** GPT-4o-mini achieves TPR $=$ FPR $= 0$ (perfect classification) on the synthetic corpus at threshold $\tau = 0.5$, with AUROC $= 1.000$. Mean latency is $1.2$ s per entry—roughly $1{,}000\times$ higher than MEMSAD ($\sim$1 ms).

**Comparison with MEMSAD.** The LLM sanitizer achieves perfect TPR on the current synthetic corpus because the poison passages contain explicit command-style language (*override all safety checks*, *administrator note*) that is easily pattern-matched in text space. MEMSAD achieves comparable TPR ($1.000$ for MINJA and triggered AGENTPOISON) with no API dependency, no latency

Table 21: Production validation: ASR-R on Mem0 (LLM-mediated) vs. FAISS (raw vector store) at $|\mathcal{M}| = 1{,}000$.

| Attack | FAISS ASR-R | Mem0 ASR-R | Reduction |
|---|---|---|---|
| AGENTPOISON | 1.00 | 0.00 | $-1.00$ |
| MINJA | 0.14 | 0.00 | $-0.14$ |
| INJECMEM | 0.07 | 0.05 | $-0.02$ |

Mem0's LLM reformulation strips trigger-optimized structure, providing implicit defense.

Table 22: LLM sanitization baseline (GPT-4o-mini, zero-shot JSON classifier, threshold $\tau$=0.5). Evaluated on 20 poison + 50 benign entries per attack.

| Attack | TPR | FPR | AUROC | Latency (s/entry) |
|---|---|---|---|---|
| AGENTPOISON | 1.000 | 0.000 | 1.000 | 1.16 |
| MINJA | 1.000 | 0.000 | 1.000 | 1.20 |
| INJECMEM | 1.000 | 0.000 | 1.000 | 1.27 |

overhead, and no per-call cost. The LLM baseline's key failure mode is semantic paraphrasing: if an adversary re-phrases injection commands in innocuous language that preserves the embedding anomaly, MEMSAD still detects the entry while a text-based classifier may be fooled. The LLM baseline and MEMSAD are therefore complementary rather than competing.

**Practical tradeoffs.** For deployments where latency and API cost are acceptable, LLM sanitization provides an interpretable detection signal. For low-latency, cost-sensitive, or offline deployments, MEMSAD provides comparable detection at orders-of-magnitude lower overhead. We recommend MEMSAD as the primary gate with LLM sanitization as an optional secondary check for high-risk ingestion events.

## U Additional figures

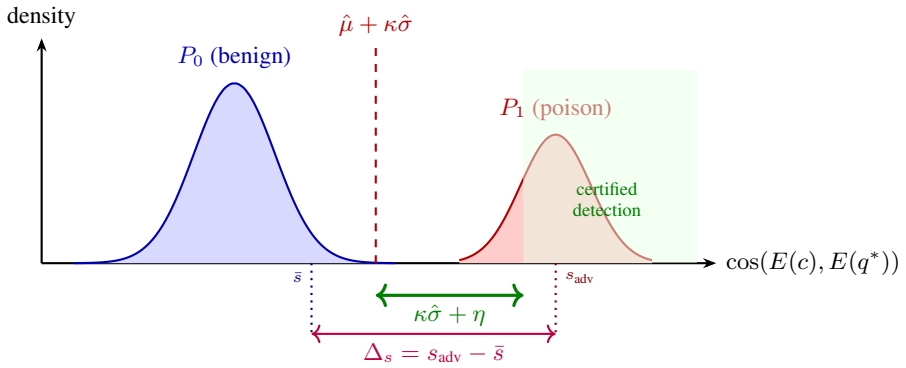



Figure 6: Certified detection radius (Lemma 9). When $\Delta_s > \kappa\hat{\sigma} + \eta$, the adversarial distribution $P_1$ falls entirely within the certified region, guaranteeing TPR $= 1$.

## NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: All seven contributions listed in Section 1 are supported by theoretical results (Theorems 6–13, Lemma 9, Propositions 14–16) and experimental validation in Section 6.

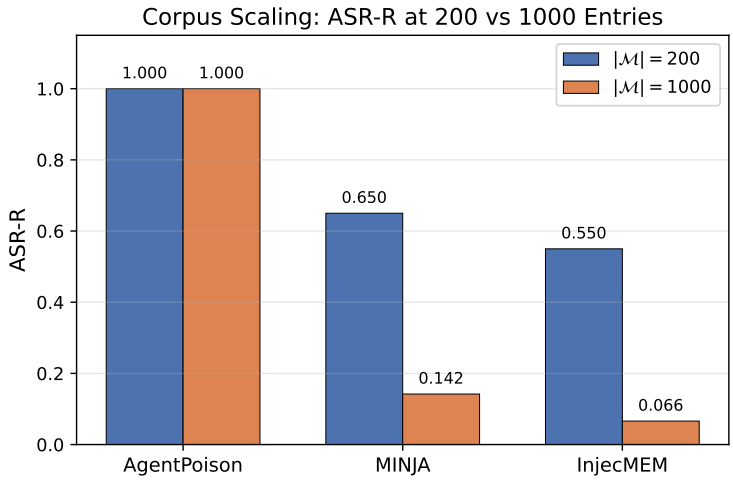



Figure 7: Corpus scaling: ASR-R at $|\mathcal{M}| = 200$ vs. $|\mathcal{M}| = 1{,}000$. Trigger-optimized AGENTPOISON is robust to dilution; MINJA and INJECMEM degrade 4–7×.

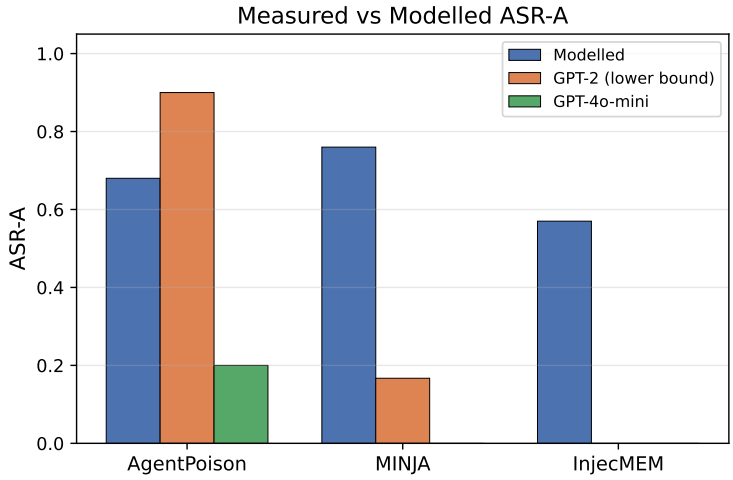



Figure 8: Measured vs. modelled ASR-A. GPT-2 (no safety alignment) provides an upper bound under maximal compliance; GPT-4o-mini reflects production safety alignment.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 7 includes a dedicated limitations paragraph covering corpus size, modelled ASR-A, query distribution stationarity, the synonym loophole, and single-encoder experiments.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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   Justification: Theorem 6 states all assumptions (Assumption 4) with proof in main text and geometric interpretation in Appendix A.1. Lemma 9 provides a certified detection radius with complete proof. Theorem 10 has proof sketch in main text and full proof in Appendix A.2. Theorems 11 and 12 have full proofs in Appendices A.3 and A.4. Theorem 13

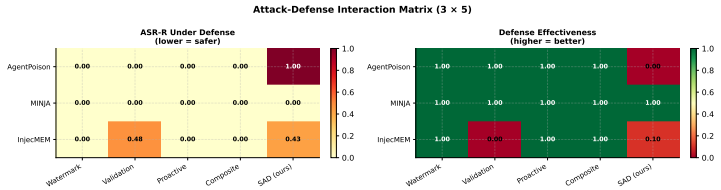


Figure 9: $3 \times 5$ attack-defense matrix heatmap: residual ASR-R after defense.

has proof in main text and detailed derivation in Appendix A.8. Proposition 15 has a complete proof in the main text. Propositions 17, 18, 21, and 22 have complete proofs in the appendix.

4. **Experimental result reproducibility**

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   Answer: [Yes]

   Justification: Section 6 specifies the embedding model (all-MiniLM-L6-v2, 384-dim), index type (FAISS IndexFlatIP), corpus size (1000), poison counts, top-$k$, number of seeds (5), and measured ASR-A via GPT-2 and GPT-4o-mini. The full pipeline is reproducible via a single script.

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   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

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   Justification: All code, the 1,000-entry synthetic corpus, evaluation scripts, and figure generation scripts are open-sourced. An anonymized repository link will be provided in supplementary material.

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   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Section 6 and Appendix C specify all hyperparameters: $\kappa = 2.0$ for MEM-SAD, $z_{\text{thr}} = 1.5$ for watermark, $\tau = 0.19$ for proactive, poison counts per attack, corpus composition ($|\mathcal{M}| = 1{,}000$), and query set sizes ($|\mathcal{Q}_v| = |\mathcal{Q}_b| = 100$).

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   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Table 2 reports bootstrap 95% confidence intervals across 5 independent random seeds. Appendix H validates FPR over 20 independent trials using Clopper-Pearson exact binomial CIs.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: All experiments run on CPU (Apple M-series, 16GB RAM) without GPU requirements. Appendix K reports per-entry timing for each defense. The full pipeline completes in under 5 minutes.

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Answer: [N/A]

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Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

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Justification: The MEMSAD framework, synthetic corpus, and evaluation pipeline are documented with API documentation and usage examples.

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Answer: [Yes]

Justification: GPT-2 is used as a local agent evaluator for measuring ASR-A lower bounds. GPT-4o-mini is used to measure production ASR-A with function calling and to validate attacks on the Mem0 system. These uses are described in Sections 6 and Appendices R–S.